



User Plane Function Setup Guide

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Affirmed Networks, Inc

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User Plane Function Setup Guide
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Preface

This document describes how to set up and configure the UPF federation in UnityCloud.

Audience

This guide is intended for all users who work with Affirmed Networks 5G or CNI products.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menu	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management
Variable parameters for which you supply values	<courier bold italic>	configure_dr --host <host name>
User input	Courier bold font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular font	result success
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.		Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<host name>
Asterisk after a CLI object indicates a required object	*	name*

Description	Convention	Examples
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>User Plane Function Setup Guide</i> . <i>Workflow</i> is an application running on the network element.
Images in online help		Many images can be enlarged by clicking the image with the left-mouse button. Alternately, they can be enlarged/reduced by pressing the Enter key when the image is highlighted (surrounded by a thick border) on the screen.

Note: Additional information that may apply to the subject text.

Tip: Helpful suggestions.

Caution: Proceed carefully to avoid possible equipment damage or data loss.

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Setting Up UPF Federation

Use the following sections to deploy UPF:

- [Prerequisites for UPF Deployment](#)
- [Configuration Requirements for UPF Deployment](#)

UPF Deployment Using NodeAffinity

UPF worker nodes are divided into two types:

- Dataplane Nodes
 - Pods that are mainly responsible for Session management and Packet forwarding.
 - Optionally, other pods that are not in the critical path of Session management and Packet forwarding like Kibana, Prometheus are anchored in the dataplane nodes.
 - Ideally, these pods are in the Non-dataplane nodes, so that more cores can be assigned to that IPPE to improve throughput and capacity.
- Non-dataplane Nodes
 - Any important system level pods that can affect the overall system like fed-opa pods, Mongo DB pods, Etcd Pods
 - UPF Singleton pods
 - Any Pods that are infra structure pods

Note: Any pod that is in the path of Session Management and packet forwarding belongs to Dataplane nodes, remaining are part of the Non-dataplane nodes.

Node Affinity and Anti-Affinity Rules

IPPE pods are set with Affinity Rule and the pods required for Non-dataplane nodes are set with Anti-affinity rules.

Dataplane Nodes:

- IPPE Pod

Non-dataplane Nodes:

- fed-opa pods
- fed-server-registry pods
- fed-db-etcd pods
- fed-mongo pods
- fed-Paas-helper Pods
- fed-UPF Singletons - UPF Monitor, CfgMgr, InterfaceMgr, L3RM, Routing
- fed-istio pods
- fed-elast-alert pods

NodeAffinity Deployment

For anchoring on Dataplane nodes:

```
{{- if eq .Values.global.deployment.nodeDataplaneAffinity
"true" }}

    affinity:

        nodeAffinity:

            requiredDuringSchedulingIgnoredDuringExecution:

                - nodeSelectorTerms:

                    matchExpressions:

                        - key: nodetype

                          operator: In

                          values: [dataplane]

                {{- end }}
```

For anchoring on Non-dataplane nodes:

```
{{- if eq .Values.global.deployment.nodeDataplaneAffinity
"true" }}

    affinity:

        nodeAffinity:

            requiredDuringSchedulingIgnoredDuringExecution:

                - nodeSelectorTerms:

                    matchExpressions:

                        - key: nodetype

                          operator: NotIn

                          values: [dataplane]

                {{- end }}
```

Note: The above NodeAffinity rules can be enabled through Helm (nodeDataplaneAffinity). Default value is **Disabled**.

Prerequisites for UPF Deployment

A UPF deployment requires:

- Cluster with SR-IOV interfaces and minimum of 16 CPUs/node and 32 GB of memory
- A minimum of two worker nodes
- Support for hugepages
- UPF Federation installation. For the full list of OpenStack / Kubernetes / SR-IOV configuration requirements, and UPF federation installation sequence and installation procedure, refer to the respective sections in the *UnityCloud Deployment Guide*.

Deploying Multus and the SRIOV Device Plugin on Affirmed VM/K8s Cluster

This section provides information on deploying Multus on K8s cluster.

Prerequisite

The UPF depends on Multus being deployed. Enable Multus as part of the **fed-sriovdp** helm chart deployment.

Downloading Multus

***Note:** Affirmed Networks does not distribute the Multus images.*

For deploying **fed-sriovdp**, download the following image to the deployment image repository: **nfvpe/multus:v3.7**

Following section is an example to download Multus when OPM is the deployment image repository:

1. Download the images on the Docker server.

```
# docker pull nfvpe/multus:v3.7
```

```
v3.7: Pulling from nfvpe/multus ab5ef0e58194: Pull complete
de03f0fc6681: Pull complete 32c03e241c62: Pull complete
Digest:
sha256:167722b954355361bd69829466f27172b871dbdbf86b85a958163
62885dc0aba
Status: Downloaded newer image for nfvpe/multus:v3.7
docker.io/nfvpe/multus:v3.7
```

- ```
docker save nfvpe/multus:v3.7 > multus_v3_7.tar
scp multus_v3_7.tar root@<opm ip>:/vnfmgr/dreg
```
- Log in to the OPM via SSH and upload the images to the repository.

```
ssh root@<opm ip>

docker load -i /vnfmgr/dreg/multus_v3_7.tar

Loaded image: nfvpe/multus:v3.7

docker images nfvpe/multus:v3.7
```

| REPOSITORY   | TAG  | IMAGE ID     | CREATED      | SIZE  |
|--------------|------|--------------|--------------|-------|
| nfvpe/multus | v3.7 | bf33fe4c52f0 | 2 months ago | 276MB |

```
docker tag <image id> <opm ip>:5005/nfvpe/multus:v3.7
docker push <opm ip>:5005/nfvpe/multus:v3.7

The push refers to repository [<opm ip>:5005/nfvpe/multus]
f2775074c27d: Pushed 0127381001ac: Pushed 77b174a6a187: Pushed
v3.7: digest:

sha256:e76725e73d9852ec99f9e16e836a54c6e89ed87275088e0180d0d
8b1c94a1d6d size: 949
```
  - Update the image location in Multus helm deployment:

```
<opm ip>:5005/nfvpe/multus:v3.7
```

The UPF depends on the SRIOV device plugin being deployed. Enable the SRIOV device plugin as part of the fed-sriovdp helm chart deployment.

## Downloading the SRIOV Device Plugin

**Note:** Affirmed Networks does not distribute the SRIOV device plugin images.

For deploying fed-sriovdp, download the following image to the deployment image repository:

**nfvpe/sriov-device-plugin:v3.3.2**

Following section is an example to download the SRIOV device plugin when OPM is the deployment image repository:

- Download the images on the Docker server.

```
docker pull nfvpe/sriov-device-plugin:v3.3.2

[root@cni-opm-163-64-25 ~]# docker pull nfvpe/sriov-device-plugin:v3.3.2

v3.3.2: Pulling from nfvpe/sriov-device-plugin
339de151aab4: Already exists
```

```

11831c082d2a: Pull complete
1161fc769add: Pull complete
508cd74d8af3: Pull complete
f4cff944bd2b: Pull complete

Digest:
sha256:bf1781254a35af473b6ba97643410537222808a89b7b9f74c2ea6da
2883f8f34

Status: Downloaded newer image for nfvp/sriov-device-
plugin:v3.3.2

docker.io/nfvpe/sriov-device-plugin:v3.3.2

docker save nfvp/sriov-device-plugin:v3.3.2 > sriov_device_
plugin_v3_3_2.tar

scp sriov_device_plugin_v3_3_2.tar root@<opm ip>:/vnfmgr/dreg

```

2. Log in to the OPM via SSH and upload the images to the repository.

```

ssh root@<opm ip>

docker load -i /vnfmgr/dreg/sriov_device_plugin_v3_3_2.tar

Loaded image: nfvp/sriov-device-plugin:v3.3.2

docker images nfvp/sriov-device-plugin:v3.3.2

```

| REPOSITORY               | TAG    | IMAGE ID     | CREATED      |
|--------------------------|--------|--------------|--------------|
| nfvp/sriov-device-plugin | v3.3.2 | e4155eb342a3 | 2 months ago |

```

42.7MB

docker tag <image id> <opm ip>:5005/nfvpe/sriov-device-
plugin:v3.3.2

docker push <opm ip>:5005 nfvp/sriov-device-plugin:v3.3.2

The push refers to repository [<opm ip>:5005/nfvpe/sriov-
device-plugin] f2775074c27d:

662fe29d2d54: Layer already exists
05e4779e81b1: Layer already exists
6f058bbe87b1: Layer already exists
f90e62552e2f: Layer already exists
32f366d666a5: Layer already exists

latest: digest:
sha256:bf1781254a35af473b6ba97643410537222808a89b7b9f74c2ea6da
2883f8f34 size: 1366

```

3. Update the image location in the SRIOV device plugin helm deployment:  
`<opm ip>:5005/nfvpe/sriov-device-plugin:v3.3.2`

## Deploying fed-sriovdp

Edit the following fed-sriovdp `<values.yaml>` file as:

- Set the `global:registry_docker_path` to the deployment image repository.
- Set the `pod-sriovdp:multus3partyAmd64Image` (for example: `nfvpe/multus:v3.7`) for the Multus AMD64 image name.
- Set the `pod-sriovdp:sriovdp3partyImage: /nfvpe/sriov-device-plugin:v3.3.2`.
- Edit the `sriovdpConfigMap` as follows:
  1. The `pod-sriovdp: sriovdpConfigMap` is used to configure the SRIOV device plugin to use the host specific PF(s).
  2. The SRIOV device plugin configuration parameters are explained at:  
<https://github.com/k8snetworkplumbingwg/sriov-network-device-plugin#configurations>
  3. The `sriovdpConfigMap` contains a `resourceList` of resourceNames. Each name corresponds to a SRIOV interface label configured in the OPM.
  4. Each `resourceName` has a list of selectors. These selectors are used to select VFs for resource pools.
  5. Due to limitations of the SRIOV device plugin, for VM deployments use the selector `pciAddresses`. The `pciAddresses` must be obtained from the VFs in use by the worker nodes.

Example for finding VFs for a VM deployment using Fortville PF:

Log into any worker node and at the prompt type:

```
%lspci
```

```
....
```

```
00:06.0 Ethernet controller: Intel Corporation Ethernet Virtual
Function 700 Series (rev 02)
```

```
00:07.0 Ethernet controller: Intel Corporation Ethernet Virtual
Function 700 Series (rev 02)
```

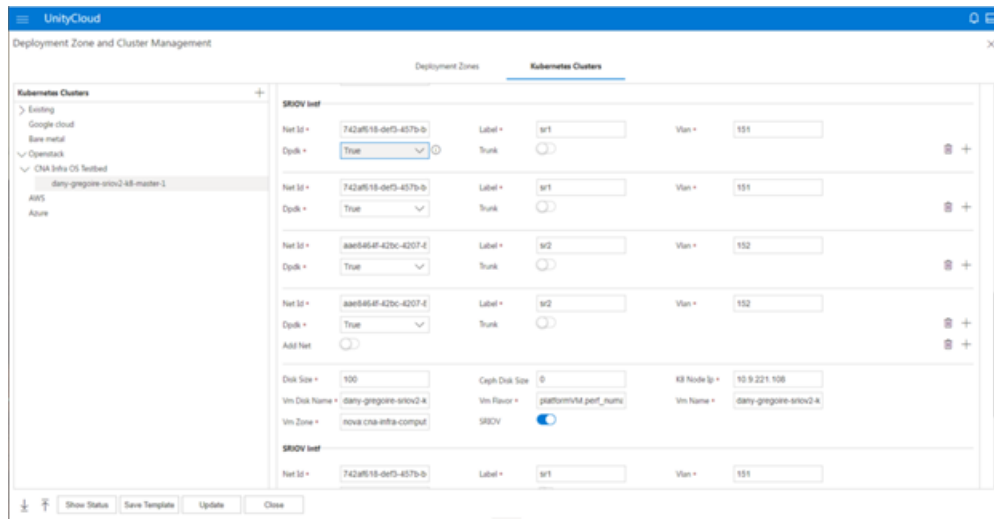
```
00:08.0 Ethernet controller: Intel Corporation Ethernet Virtual
Function 700 Series (rev 02)
```

```
00:09.0 Ethernet controller: Intel Corporation Ethernet Virtual
Function 700 Series (rev 02)
```

```
....
```

The order of the `lspci` listing is guaranteed to match the order of each node's **SRIOV Intf** listing as provisioned in the OPM.

**Figure 1-1 Deployment Zone and Cluster Management**



This example shows that the first two PCI listed in the **lspci** output belongs to the label **sr1**; these are the first two entries provisioned in the OPM. The last two listed PCI belongs to the label **sr2**.

The OPM configuration offers the flexibility to provision the SRIOV interfaces for each node. However, there is only one configmap for the SRIOV device plugin. Each node on the OPM must configure the SRIOV Interfaces using the same sequence. This ensures that the same configmap applies to each node.

The list of PCI addresses can then be used to create the configmap for the SRIOV device plugin.

Example for VM node deployment using Fortville PF:

```
Example VM w/ fvl PF

sriovdpConfigMap:

 resourceList:

 - resourceName: "sr1"

 selectors:

 drivers:

 - "vfio-pci"

 pciAddresses:

 - "0000:00:06.0"

 - "0000:00:07.0"

 - resourceName: "sr2"

 selectors:

 drivers:
```

```

- "vfio-pci"
pciAddresses:
- "0000:00:08.0"
- "0000:00:09.0"

```

For Mellanox, the **sriovdpConfigMap** is the same except for the driver that is **mlx5\_core**.

## Verifying SRIOV Device Plugin Configuration

To verify the SRIOV device plugin interfaces configuration:

1. Check if the **kube-sriov-device-plugin-amd64-xxx** related pods are running on the master node and the pod itself.

**Example:**

```

$ kcp --all-namespaces ...

kube-system kube-sriov-device-plugin-amd64-4qjn8 1/1
Running 0 3d20h

kube-system kube-sriov-device-plugin-amd64-7j8pw 1/1
Running 0 3d20h

kube-system kube-sriov-device-plugin-amd64-rrwfl 1/1
Running 0 3d20h

```

2. Log in to the PPE pod and execute the following command to verify the interfaces are configured and running:

**Note:** Ensure that the SRIOV network labels are visible in the PPE pod.

**Example:**

```

$ kubectl get nodes

NAME STATUS ROLES AGE VERSION
hostname-k8-master-1-cca076-4b2c09f7b029c1b9 Ready master 3d22h v1.18.3
hostname-node-1-cca076-4b2c09f7b029c1b9 Ready <none> 3d22h v1.18.3
hostname-k8-node-2-cca076-4b2c09f7b029c1b9 Ready <none> 3d22h v1.18.3

$ kubectl describe hostname-k8-node-1... | grep sr
...

```

```
dataplane.io/sr1 1 1
dataplane.io/sr2 1 1
...
```

## Network Attachments

Network Attachments are created for SRIOV interfaces for the following:

- Mellanox PFs.
- Fortville PFs that optionally use the kernel driver, as opposed to DPDK (vfio).

### Creating Network Attachments

Network attachments are created by editing the **values.yaml** file and adding **pod-cdpln:sriovNetworks**. An list entry is added for each SRIOV interface that requires a network attachment.

#### Example:

```
pod-cdpln:
sriovNetworks:
 - name: "sriov-net1"
 labels: "sr1"
 - name: "sriov-net2"
 labels: "sr2"
```

The **name** field is the name of the sriov network attachment definition to be created.

The **label** field is the SRIOV-DP label applied to this network, either at OPM cluster creation time or commissioning time.

### Enabling/Disabling Network Attachments

To enable network attachment capability, modify the pod-ppe HELM chart.

**Note:** While deploying, modify the actual **.yaml** file.

To annotate the deployment files and to specify the network attachments in the interface directory of the pod-ppe repo, edit the **values.yaml** file and set the pod-ppe attribute **netAttachDefs** to **enabled:true**.

```
value.yaml # pod-ppe:
...
netAttachDefs:
 enabled : true
```

```
annotation:
"sriov-net1,
sriov-net2"
```



## Deploying Mellanox

To deploy UPF in the Mellanox environment:

- [Enabling/Disabling Mellanox Capability](#)
- [Verifying Mellanox Configuration](#)

### Enabling/Disabling Mellanox Capability

To enable Mellanox capability:

1. Create network attachments for each Mellanox PF.
2. Enable the network attachment annotations in the deployment.

***Note:** For creating and enabling network attachments, please refer to [Network Attachments](#) section.*

### Verifying Mellanox Configuration

To verify the Mellanox interfaces configuration:

1. Check if the multus related pods are running on the master node and the pod itself.

Example:

```
$ kcp --all-namespaces
```

```
...
```

```
kube-system kube-multus-ds-amd64-jfbcg 1/1 Running 0 88s
```

```
kube-system kube-multus-ds-amd64-vnqxs 1/1 Running 0 88s
```

```
kube-system kube-multus-ds-amd64-zrzs5 1/1 Running 0 88s
```

```
...
```

2. Log in to the PPE pod and execute the following command to verify the interfaces are configured and running:

***Note:** Ensure that the **net1** and **net2** interfaces are visible in the PPE pod.*

Example:

```
$ kubectl exec -it ppe-1-0-0-380-main-int-dbg-54f88d4dcd-ph5ss
```

```
bash -n fed-upf
```

```
ifconfig -a
```

```
...
```

```

net1: flags=4098<BROADCAST,MULTICAST> mtu 1500
ether fa:16:3e:e8:ae:75 txqueuelen 1000 (Ethernet)
RX packets 35 bytes 3230 (3.1 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
net2: flags=4098<BROADCAST,MULTICAST> mtu 1500
ether fa:16:3e:2b:0c:1c txqueuelen 1000 (Ethernet)
RX packets 1 bytes 64 (64.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
...

```

## IPPE/TDEF Deployments of the UPF

For IPPE / TDEF deployments of the UPF, **Guaranteed Scheduling** is enabled under both **ppe** and **tdef** sections of the chart.

***Note:** Specify the resource settings (CPU/Memory values) for the individual containers in the **Guaranteed Scheduling** section.*

For the PPE or IPPE deployment in a UPF –

- When **Guaranteed Scheduling** is disabled, the values in the **resources** section are considered.
- When **Guaranteed Scheduling** is enabled, the values specified in the **Guaranteed Scheduling** section are considered.

***Note:** For OPM based deployments, these options are available on the **install federation** screen when you enable **display all parameters**.*

**Example:**

**Note:** Specify SR-IOV VF label for Proxy container.

In earlier releases, the label name was specified as **SR1**. From 1.4.2 onwards, any label name can be used.

SR-IOV VF label must be from the same PF as PPE VF.

#### When **Guaranteed Scheduling** enabled –

```
pod-ppe:
 clusterRole: permissive-full-cr
 deployment:
 deployCaptureAgent: false
 podAntiAffinityEnabled: false
 replicas: 1
 enableJumboFrames: false
 enabled: true
 externalSvc:
 enabled: false
 masterNodeIpAddress: 10.31.49.228
 guaranteedScheduling:
 enabled: true
 infraCpuLimit: 100m
 infraMemoryLimit: 256Mi
 kafkaCpuLimit: 100m
 kafkaMemoryLimit: 256Mi
 ppeCpuLimit: 3
 ppeMemoryLimit: 8Gi
 proxyCPULimit: 200m
 proxyInitCPULimit: 100m
 proxyInitMemoryLimit: 128Mi
 proxyMemoryLimit: 256Mi
 netAttachDefs:
 annotation: sriov-net1, sriov-net2
 enabled: false
 ppeFirstWorker: "1"
 ppeIoScaling: separated-cores
 ppeNumWorkers: "2"
 rbacEnabled: true
 resources:
 hugepagesType: 2Mi
 ppeCpuLimits: 3
 ppeCpuRequests: 3
 ppeHugepagesSize: 4Gi
 ppeMemory: 8Gi
 serviceAccountName: ppe-serviceaccount
 sessionPersistenceEnabled: false
 sriov:
 - labels: srl
 nbIfc: "1"
```

```

- labels: sr2
 nbIfc: "1"
tdef:
 enabled: true
 name: ippe
 proxy:
 guaranteedScheduling:
 enabled: true
 tdefProxyCpuLimit: 4
 tdefProxyTaskMemoryLimit: 4Gi
 resources:
 hugepagesType: 2Mi
 proxyCpuLimits: 3
 proxyCpuRequests: 3
 proxyHugepagesSize: 4Gi
 proxyMemory: 8Gi

 sriov:
 - labels: sr1
 nbIfc: "1"

```

**Note:**

Number of proxy task is calculated as *tdefProxyCpuLimit-1*.

The recommended *tdefProxyMemoryLimit* is 4Gi.

Proxy container memory limit is calculated as  $(tdefProxyCpuLimit-1) * tdefProxyMemoryLimit$ .

## Configuration Requirements for UPF Deployment

After UPF federation installation, configure the interfaces for N3 and N6, PFCP profiles, UE pools, and Static Routes for complete UPF deployment.

Following sections provide detailed instructions on:

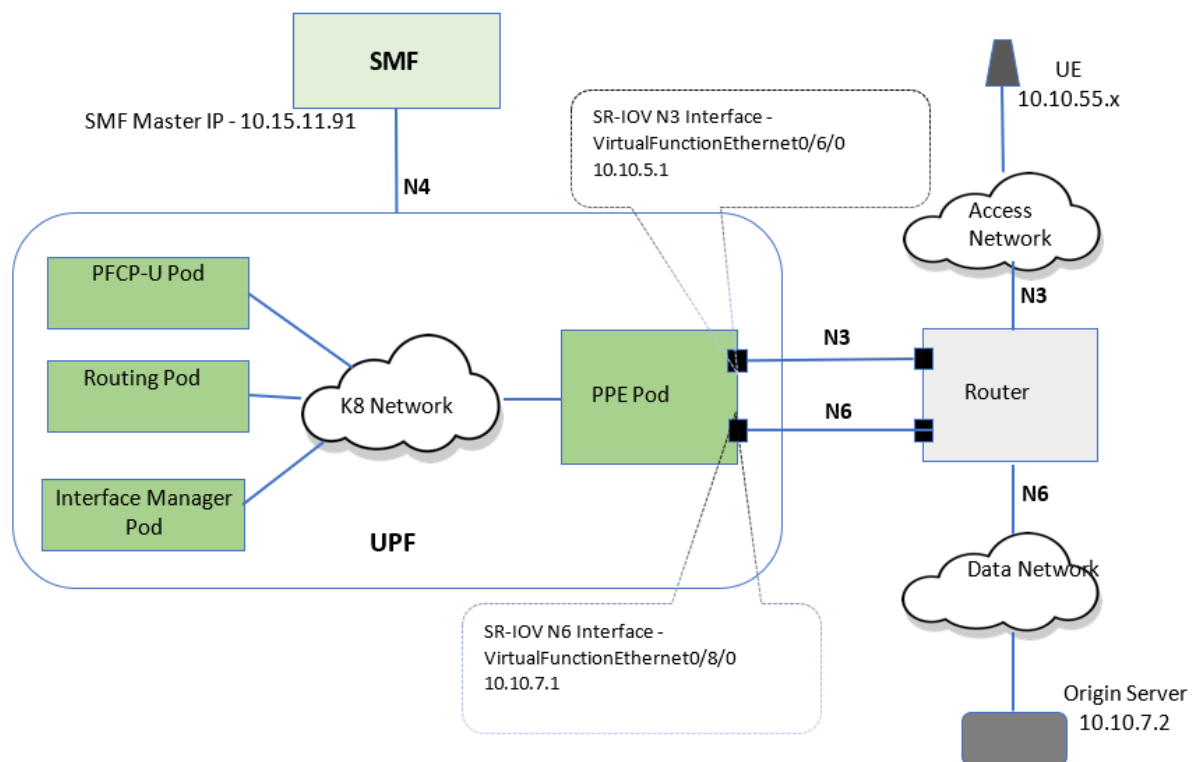
- [CLI Support - Configuration of Interfaces with VLAN](#)
- [Interface Management – Shared Subnet Network Design](#)
- [Interface Management – Non-overlapping Subnet Network Design](#)
- [Oper CLI Support for Interface Configuration](#)
- [CLI Support - Configuration of UE IP Pools](#)
- [Oper CLI Support for UE IP Pools](#)
- [PFCP Association Management](#)
- [CLI Support - Configuration of BFD Profiles](#)

- CLI Support - Configuration of Static Routes
- CLI Support - Enabling BFD for Static Routes
- Oper CLI Support for Static Routes
- Dynamic Routing
- NRF Registration

## Reference Diagram

The following architecture depicts configuration parameters for UPF deployment.

**Figure 1-2 UPF Setup - Reference Diagram**



## CLI Support - Configuration of Interfaces with VLAN

Use the following steps to configure the interface templates to manage the SR-IOV interfaces associated with the PPE pods:

1. Configure the **Network Context** where the interfaces are anchored.
2. Configure the VLAN and the address specific template for any interfaces that are detected on the PPE pods – a free subnet is assigned to each interface as configured in the template.
3. Execute the following commands:

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:10:42(config)# interface-manager network-context TEST
```

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:11:03(config-network-context-TEST)# commit
```

Commit complete

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:11:16(config)# interface-manager network-context TEST ip-
interface intf-100
```

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:11:41(config-ip-interface-intf-100)# commit
```

Commit complete.

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:11:48(config)# interface-manager network-context TEST
ip-interface intf-100 vlan-id 100 ipv4-address-template ANET_
IFV4 base-address 10.10.10.10 count 2 prefix-length 24
```

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:12:55(config-ipv4-address-template-ANET_IFV4)# commit
```

Commit complete.

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:13:10# show running-config interface-manager network-context
TEST
```

```
interface-manager
network-context TEST
ip-interface intf-100
vlan-id 100
ip-mtu 1500
ipv4-address-template ANET_IFV4
base-address 10.10.10.10
prefix-length 24
count 2
!
!
```

!

!

# Interface Management

## Overview

Interface Manager is responsible for configuring and distributing interfaces in a UPF federation. It is deployed as singleton (single pod) deployment and supports both traditional **non-overlapping** and **shared-subnet** deployments. It handles configuration processing of interfaces for all network-instances and creates a globally unique interface name/id for each interface generated by configuration templates. It is used as global identifier for interfaces and has a defined scheme for interface naming to accommodate all interface types.

## Configuration

The following sections provide detailed information on CLI support for:

- [CLI Support - Configuration of Interfaces with VLAN](#)
- [Interface Management – Shared Subnet Network Design](#)
- [Interface Management – Non-overlapping Subnet Network Design](#)
- [Oper CLI Support for Interface Configuration](#)

## Statistics

For more information on Statistics, see [Interface Manager Statistics Dashboard](#).

## Interface Management – Shared Subnet Network Design

This section describes shared subnet network design.

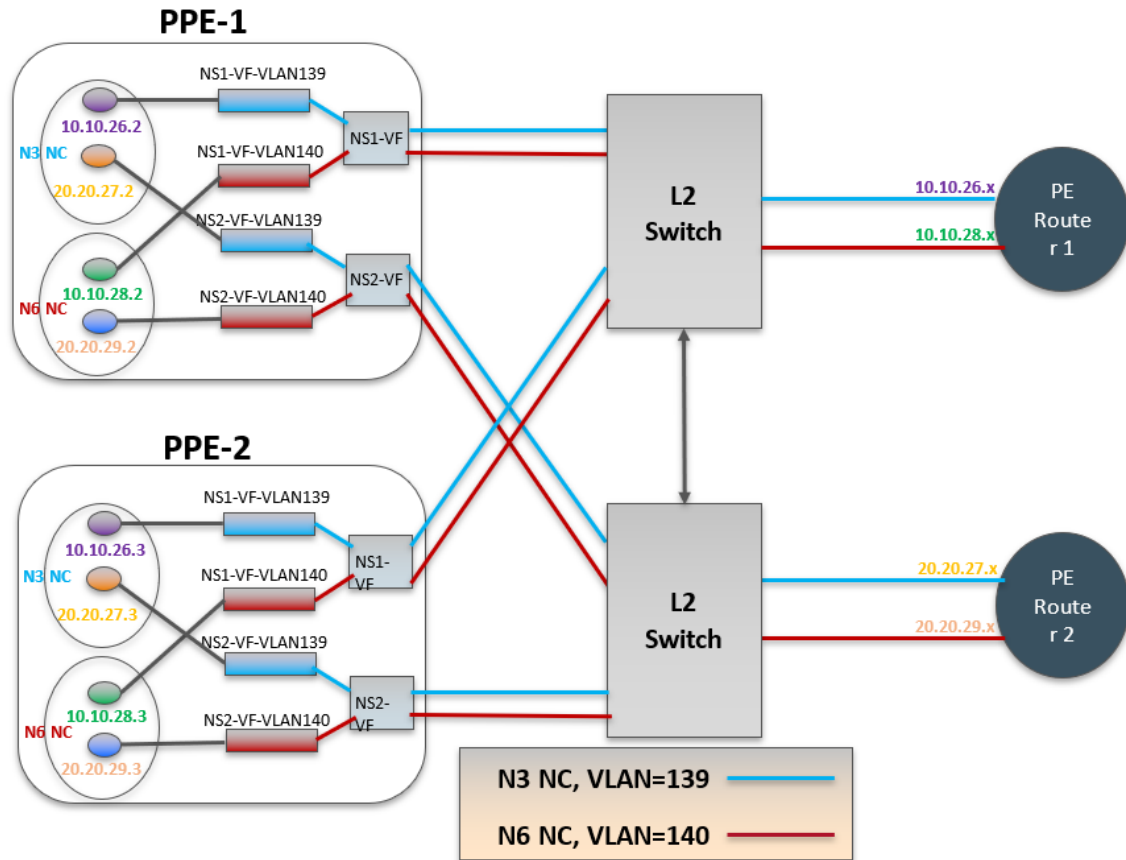
- Every interface on a PPE instance must be on a separate subnet. Interfaces across PPE instances can be on the same shared subnet.
- Every PPE interface has the same number of interfaces and the connectivity from every PPE instance is identical. (that is, no single PPE instance contains special interface configuration requirements)
- For every interface on a PPE instance that is on a shared-subnet, a “ipv4(or ipv6)\_ address template” object must be configured.
- The address template specifies the starting address of the shared-subnet to use and the number of addresses that are assigned on this subnet.
- Depending on the number of interfaces on a PPE instance, an equal number of address templates are configured for a given interface. For example, if each PPE has two interfaces (VF), then each of the sub-interfaces configured (for a given NC and VLAN) has two address templates on two different subnets. This results in individual addresses being assigned to both of the interfaces for every PPE instance that is created.



- When a new PPE instance is created, the address templates result in generated config with a specific address assigned to the interfaces on this PPE instance.

## Example for Shared Subnet Interface Management

Figure 1-3 Shared Subnet Interface Management



## Example for Shared Subnet Interface Manager Configuration

```
interface-manager
network-context N3
ip-interface intf-139
vlan-id 139
ip-mtu 1500
shared-subnet true
ipv4-address-template N3_IPv4_INTF1
base-address 10.10.26.2
prefix-length 24
count 4
```

```

!
ipv4-address-template N3_IPv4_INTF2
 base-address 20.20.27.2
 prefix-length 24
 count 4
!
!
!
network-context N6
 ip-interface intf-140
 vlan-id 140
 ip-mtu 1500
 shared-subnet true
 ipv4-address-template N6_IPv4_INTF1
 base-address 10.10.28.2
 prefix-length 24
 count 4
 !
 ipv4-address-template N6_IPv4_INTF2
 base-address 20.20.29.2
 prefix-length 24
 count 4
 !
!
!
!
!

```

## Interface Management – Non-overlapping Subnet Network Design

This section describes the steps to configure the interface templates to manage the SR-IOV interfaces associated with the PPE pods.

- Configure the Network Context where the interfaces are anchored.

- Configure the VLAN and the address specific template for interfaces detected on the PPE pods – a free subnet is assigned to each interface as configured in the template.
- In this design, if every sub-interface (associated with VLAN/NC) is assigned an IPv4 subnet, an IPv4 address template is configured with the base-address and the number of subnets that is assigned.

### **interface-manager network-context TEST**

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:10:42 (config) # interface-manager network-context TEST

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:11:03 (config-network-context-TEST) # commit

Commit complete.
```

### **interface-manager network-context TEST ip-interface intf-100**

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:11:16 (config) # interface-manager network-context TEST
ip-interface intf-100 vlan-id 100

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:11:41 (config-ip-interface-intf-100) # commit

Commit complete.
```

### **interface-manager network-context TEST ip-interface intf-100 vlan-id 100 ipv4-address-template ATT\_IFV4 base-address 10.10.10.10 count 2 prefix-length 2**

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:11:48 (config) # interface-manager network-context TEST
ip-interface intf-100 ipv4-address-template ATT_IFV4 base-
address 10.10.10.10 count 2 prefix-length 24

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:12:55 (config-ipv4-address-template-ATT_IFV4) # commit

Commit complete.
```

### **show running-config interface-manager network-context TEST**

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:13:10 # show running-config interface-manager network-context
TEST

interface-manager
network-context TEST

 ip-interface intf-100
 vlan-id 100
```

```

ip-mtu 1500

ipv4-address-template ATT_IFV4
 base-address 10.10.10.10
 prefix-length 24
 count 2
!
!
!
!

```

## Oper CLI Support for Interface Configuration

Execute the following command:

```

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:13:19# show interface-manager v1 provisioned interface

{
 "InterfaceInstances": [

 {
 "if-id": 5,
 "ipv4-addresses": [
 {
 "ipv4": "10.10.10.10",
 "prefix-length": 24
 }
],
 "l3-mtu": 1500,
 "network-context": "TEST",
 "vlan-id": 100
 },
 {

```

```
"if-id":6,
"ipv4-addresses":[
 {
 "ipv4":"10.10.11.10",
 "prefix-length":24
 }
],
"l3-mtu":1500,
"network-context":"TEST",
"vlan-id":100
}
]
}

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-67c6b97b47-7mlvv
08:15:05#
```

# Loopback Management

This section provides details on Loopback Management functions:

- Allows configuration of the loopback templates - with multiple ranges of {Start, End} address pairs with explicit start and end range in addition to prefix-length.
- Supports IPv4 and IPv6 loopbacks.
- Supports the scope for loopbacks - global, pod, and dynamic.

## Configuration

### Pod scope loopback

```
admin@pod-cfgmgr-bbc7d4956-njrr8 07:27:05# show running-config
interface-manager network-context N3 loopback-ipv4-interface
n3Lpbk1
```

```
interface-manager
```

```
network-context N3
```

```
loopback-ipv4-interface n3Lpbk1
```

```
loopback-start-address 192.1.1.1
```

```
loopback-end-address 192.1.1.10
```

```
admin-state enabled
```

```
loopback-scope pod
```

```
loopbacks-per-pod 1
```

```
tdef false
```

```
route-tag 65535
```

```
route-tag-aggregate 65535
```

```
route-summary-prefix-length 32
```

```
advertise-aggregate true
```

```
admin@pod-cfgmgr-bbc7d4956-njrr8 07:27:20# show running-config
interface-manager network-context N3 loopback-ipv6-interface
n3Lpbk1_v6
```

```
interface-manager
```

```
network-context N3
```

```
loopback-ipv6-interface n3Lpbk1_v6
```

```
loopback-start-address 4192:1:1:1:1:1:1:1
```

```

loopback-end-address 4192:1:1:1:1:1:1:a
admin-state enabled
loopback-scope pod
loopbacks-per-pod 1
tdef false
route-tag 65535
route-tag-aggregate 65535
route-summary-prefix-length 128
advertise-aggregate true
!
!
!
admin@pod-cfgmgr-bbc7d4956-njrr8 07:30:06#

```

## Global scope

```

admin@pod-cfgmgr-bbc7d4956-njrr8 07:41:28# show running-config
interface-manager network-context N6 loopback-ipv4-interface
n6-loopback-intf

loopback-ipv4-interface n6-loopback-intf

loopback-start-address 1.2.3.4
loopback-end-address 1.2.3.4
admin-state enabled
loopback-scope global

admin@pod-cfgmgr-bbc7d4956-njrr8 07:44:40# show running-config
interface-manager network-context N6 n6-loopback-intf_v6

loopback-ipv4-interface n6-loopback-intf_v6

loopback-start-address 4195:1:1:1:1:1:1:1
loopback-end-address 4195:1:1:1:1:1:1:10
admin-state enabled
loopback-scope global

```

## Dynamic Scope

```
admin@pod-cfgmgr-bbc7d4956-njrr8 07:46:44# show running-config
interface-manager network-context N3 loopback-ipv4-interface
n3Loop1

interface-manager

network-context N3

loopback-ipv4-interface n3Loop1

loopback-start-address 12.1.1.1
loopback-end-address 12.1.1.10
admin-state enabled
loopback-scope dynamic
route-tag 65535
route-tag-aggregate 65535
route-summary-prefix-length 32
advertise-aggregate true

admin@pod-cfgmgr-bbc7d4956-njrr8 07:46:45# show running-config
interface-manager network-context N3 loopback-ipv4-interface
n3Loop1_v6

interface-manager

network-context N3

loopback-ipv4-interface n3Loop1

loopback-start-address 3194:1:1:1:1:1:1:1
loopback-end-address 3194:1:1:1:1:1:1:10
admin-state enabled
loopback-scope dynamic
route-tag 65535
route-tag-aggregate 65535
route-summary-prefix-length 32
advertise-aggregate true.

admin@pod-cfgmgr-bbc7d4956-njrr8 07:47:08#
```



## CLI

```
admin@pod-cfgmgr-bbc7d4956-njrr8 07:49:10# show interface-
manager v1 provisioned loopback
```

```
{
 "LoopbackInstances": [
 {
 "if-id": 5,
 "ipv4-loopback": [
 {
 "ipv4": "22.1.1.1"
 }
],
 "loopback-scope": "dynamic",
 "network-context": "N3",
 "pod-type": "ppe"
 }
]
}
```

```
admin@pod-cfgmgr-bbc7d4956-njrr8 07:49:02# show interface-
manager v1 free loopback
```

```
{
 "FreeLoopbackInstances": [
 {},
 {},
 {
 "ipv4-loopback": [
 {
 "ipv4": "192.1.1.2"
 }
],
 },
 {
```

```

"ipv4-loopback":[
{
"ipv4":"193.1.1.2"
}
],
{
"ipv4-loopback":[
{
"ipv4":"194.1.1.2"
}
],
},
{
"ipv4-loopback":[
{
"ipv4":"195.1.1.2"
}
],
},
{
"ipv6-loopback":[
{
"ipv6":"4192:1:1:1:1:1:1:2"
}
],
},
{
"ipv6-loopback":[
{
"ipv6":"4193:1:1:1:1:1:1:2"
}
]
}

```

```

}
]
},
{
 "ipv6-loopback": [
 {
 "ipv6": "4194:1:1:1:1:1:1:2"
 }
]
},
{
 "ipv6-loopback": [
 {
 "ipv6": "4195:1:1:1:1:1:1:2"
 }
]
}
]
}

admin@pod-cfgmgr-bbc7d4956-njrr8 07:49:10#

```

The following section provides Loopback Management configuration:

```

show running-config interface-manager network-context N3

interface-manager
network-context N3
loopback-ipv4-interface LoopbackAddressIPv4
loopback-address 1.2.3.4
prefix-length 24
admin-state enabled
!

```

## Configuration of Loopback Template

The following section provides configuration of Loopback Template:

**Note:** While upgrading the *tdef* attribute under *pod-scope*, *loopback* must be set to **true**, if the *loopback* template is used for TDEF services.

```
admin@pod-cfgmgr-2-1-0-758-main-int-dbg-8444f9b5b-l9qrv
11:18:22 (config) # interface-manager network-context N6
loopback-ipv4-interface FOO loopback-scope pod
admin@pod-cfgmgr-2-1-0-758-main-int-dbg-8444f9b5b-l9qrv
11:18:35 (config-loopback-ipv4-interface-FOO) # ?
Possible completions:
 admin-state Administrative state of loopback
interface - optional - default is enabled
 loopback-end-address IPv4 address of loopback interface -
end address for the range
 loopback-scope Scope of loopback interface -
default is global
 loopback-start-address IPv4 address of loopback interface -
start address for the range
 loopbacks-per-pod Number of loopback interfaces
assigned to a ppe when the scope is type pod
 tdef Indicates if this loopback template
is used for TDEF services - default is false

admin@pod-cfgmgr-1-0-0-453-main-int-dbg-964cf44dc-dbqkt
05:29:47 (config-loopback-ipv4-interface-TEST1) # loopback-scope
Possible completions:

global Reachable and advertised to all dataplane pods in the
federation

pod Reachable and advertised only by the dataplane pod it is
anchored in

dynamic Dynamically anchored by another application
```

### Sample Configuration

```
admin@pod-cfgmgr-2-1-0-758-main-int-dbg-8444f9b5b-l9qrv
11:18:22# show running-config interface-manager network-context
N6 loopback-ipv4-interface
loopback-ipv4-interface BAR loopback-start-address
2.3.4.5 loopback-end-address 2.3.4.10 admin-state
enabled loopback-scope pod loopbacks-per-pod 2 tdef
```

```

false !
 loopback-ipv4-interface FOO loopback-start-address
1.2.3.4 loopback-end-address 1.2.3.4 admin-state
enabled loopback-scope global !

```

## Configuration Support for LI Profile

The port range profile specifies the range of ports assigned to each ppe when they are instantiated.

It provides the option to configure **start-port value**, **end-port value**, and **ports-per-pod**.

**Port-range** assigned to each ppe is published through etcd.

```

show running-config interface-manager port-profile
interface-manager
port-profile PORTPROF-1
start-port-range 10000
end-port-range 30000
ports-per-pod 50
!
!

show ppe v1 li resource-info ppeId <integer>

```

**Note:** This command is used to display the local loopback address and port ranges used by LiApp in a PPE.

```

admin@pod-cfgmgr-2-1-0-758-main-int-dbg-8444f9b5b-l9qrv
11:32:45(config)# upf service-construct li-profile ?
Possible completions:
 end-port End port value to be used for LI on this
UPF.
 ipv4-loopback-name Name of the ipv4 Loopback Template to be
used for LI on this UPF.
 ipv6-loopback-name Name of the ipv6 Loopback Template to be
used for LI on this UPF.
 start-port Start port value to be used for LI on
this UPF.

```

**Note:**

- Pod-Scope loopback must be configured under li-profile
- Port-Range profile configuration is not needed anymore
- Start-Port and End-Port values must be configured within the li-profile

## Sample Configuration

```
show ppe v1 li resource-info ppeId 1

{
 "loopbackIPv4Addr": "1.2.3.4",
 "loopbackIPv6Addr": "2001:adb8:a0b:12f0::102:30b",
 "port-range-end": 10049,
 "port-range-start": 10000
}

admin@pod-cfgmgr-2-1-0-758-main-int-dbg-8444f9b5b-19qrv
11:32:18 (config) # show full-configuration upf service-construct
li-profile
upf service-construct li-profile ipv4-loopback-name LI-TEST
upf service-construct li-profile start-port 1000
upf service-construct li-profile end-port 2600
```

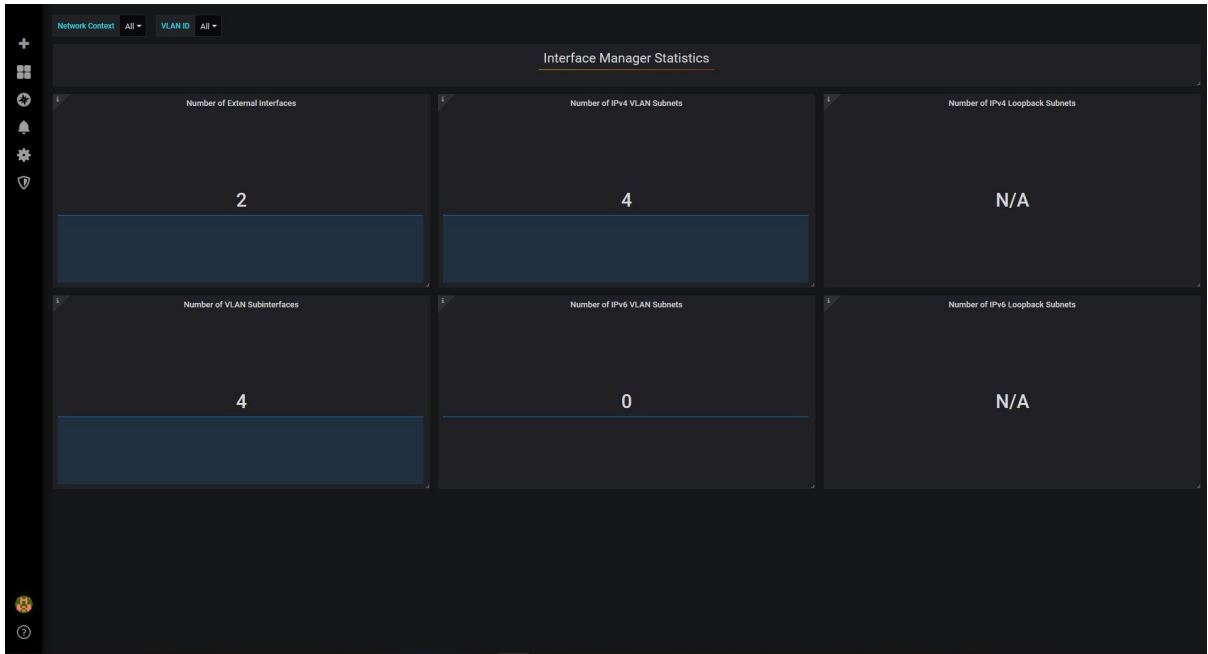
## Interface Manager Statistics Dashboard

This section provides the default Grafana dashboard view and its panel descriptions for Interface Manager Statistics.

**Dashboard Title:** Interface Manager Statistics

**Dashboard Description:** **Interface Manager Statistics** provides graphical views for the number of external interfaces, subinterfaces, IPv4/IPv6 VLAN subnets, and their loopback subnets.

**Figure 1-4 Interface Manager Statistics**



For more information on the Dashboards and Panel descriptions, see [Interface Manager Grafana Dashboards](#).

The following table provides Panel titles and its descriptions in the **Interface Manager Statistics**:

**Table 1-1 Panel Titles and its Descriptions in the Interface Manager Statistics**

| Panel Title                     | Panel Description                                                |
|---------------------------------|------------------------------------------------------------------|
| Number of External Interfaces   | This panel provides the number of external interfaces connected. |
| Number of IPv4 VLAN Subnets     | This panel provides the number of IPv4 VLAN subnets.             |
| Number of IPv4 Loopback Subnets | This panel provides the number of IPv4 loopback subnets.         |
| Number of VLAN Subinterfaces    | This panel provides the number of VLAN subinterfaces.            |
| Number of IPv6 VLAN Subnets     | This panel provides the number of IPv6 VLAN subnets.             |
| Number of IPv6 Loopback Subnets | This panel provides the number of IPv6 loopback subnets.         |

## UE IP Pool Management

This section describes UE IP pool management.

## CLI Support - Configuration of UE IP Pools

Use the following steps to configure UE Pools on the UPF:

1. Configure the Network Context (that is, the N6 Network Context).
2. Configure the UE Pool in the respective Network Context.
3. Add the Network Context under **upf service**.

To configure NC/IP-Pools, execute the following commands:

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-7db68bdc6f-x64tv
05:42:01 (config) # upf service-construct network-context <name_
of_network_context>

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-7db68bdc6f-x64tv
05:43:02 (config-network-context-N6NC) #ip-pool <name_of_ip_pool>
```

### Sample Configuration:

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-7db68bdc6f-x64tv
05:46:48# show running-config upf service-construct network-
context

upf service-construct network-context N6NC

type N6

ip-pool LOCAL

type UE-IP-Pool
range-start 1.1.1.1
range-end 1.1.1.100
route-summary-prefix-length 24
!
!

upf service-construct network-context N6_02

type N6

ip-pool LOCAL

type UE-IP-Pool
range-start 5.5.5.5
range-end 5.5.5.50
route-summary-prefix-length 24
!
```



```

!
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-7db68bdc6f-x64tv
05:47:12#

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-7db68bdc6f-x64tv
05:45:19# show running-config upf service

upf service admin-state enabled

upf service smf-serving-area SL1

upf service network-context N6NC

!

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-7db68bdc6f-x64tv
05:46:47#

```

## Oper CLI Support for UE IP Pools

To display the configured NC and UE IP Pools, execute the following command:

```

show upf v1 provisioned ip-pool

{
 "Network-Contexts": [
 {
 "IP Pools": [
 {
 "end-ip": "1.1.1.100",
 "name": "LOCAL",
 "route-summary-prefix-length": 24,
 "start-ip": "1.1.1.1",
 "type": "UE-IP-Pool"
 }
],
 "name": "N6"
 },
 {
 "IP Pools": [

```

```
{
 "end-ip": "5.5.5.50",
 "name": "LOCAL",
 "route-summary-prefix-length": 24,
 "start-ip": "5.5.5.5",
 "type": "UE-IP-Pool"
}
],
"name": "N6_02"
}
]
```

## PFCP Association Management

PFCP Association management on the UPF supports the following capabilities:

- PFCP Association is always initiated from the SMF to the UPF.
- When the UPF receives a PFCP Association request from the SMF/PFCP, it responds back with a PFCP Association Response with the TEID range to use as well as the GTP-U loopbacks to use on the N3 network context.
- When the PFCP Association is disabled, the UPF clears all the remaining sessions associated with this PFCP Association.

## Configuration

```
admin@pod-cfgmgr-5-4-0-81-rel-4-2-0-5b85684cf6-kb6n4 09:19:17#
show running-config pfc

pfc admin-state enabled

pfc association fqdn upf.affirmed.com

pfc association default-profile DEFAULTPROF

pfc pfc-message-ppe-batch-timer-interval 10

pfc pfc-message-ppe-batch-size 10

pfc peer SMF-1

fqdn smf.affirmed.com

port 8805

ipv4-address 10.144.194.41

!

pfc profile DEFAULTPROF

pfc-association-retry-interval 60000

pfc-session-response-timeout 2400

pfc-session-retransmit-count 2

pfc-node-response-timeout 2000

pfc-node-retransmit-count 3

pfc-heartbeat-periodic-interval 60000

pfc-heartbeat-down-threshold 0

pfc-guard-time-interval 60000

dscp-marking df
```

```

!
pfcf trace trace-keys-admin-state enabled
pfcf trace verbose disabled
pfcf trace probability 0.000000000000000e+0
admin@pod-cfgmgr-5-4-0-81-rel-4-2-0-5b85684cf6-kb6n4 09:23:37#

```

## CLI

```

admin@pod-cfgmgr-5-4-0-81-rel-4-2-0-5b85684cf6-kb6n4 09:19:10#
show pfcf vl association-data

```

```

{
 "AssociationInstances": [
 {
 "association-entry": {
 "association-recent-up-enter-time": "Sun Nov 24 15:47:14 2024",
 "association-status": "UP",
 "cp-features": {
 "LOAD": false,
 "MPAS": false,
 "OVRL": false
 },
 "heart-beat-status": "UP",
 "local-ip": "10.144.194.34",
 "local-node-id": "upf.affirmed.com",
 "local-recovery-value": 2701023604,
 "peer-node-id": "smf.affirmed.com",
 "peer-path-info": [
 {
 "heartbeat-recent-up-enter-time": "Sun Nov 24 15:47:14 2024",
 "path-heart-beat-status": "UP",
 "peer-ip-address": "10.144.194.41"
 }
],

```

```
"peer-recovery-value":2701023595,
"pfcsp-smf-set-id":"","
"up-features":{
 "ADPDP":false,
 "ATSSS_LL":false,
 "BUNDL":false,
 "BUPC":false,
 "CIOT":false,
 "DDDS":false,
 "DDND":true,
 "DLBD":true,
 "DPDRA":false,
 "EMPU":true,
 "EPFAR":false,
 "ETHAR":false,
 "FRRT":false,
 "FTUP":true,
 "GCOM":false,
 "GPQM":false,
 "HEEU":false,
 "IP6PL":false,
 "IPTV":false,
 "MNOP":false,
 "MPAS":true,
 "MPTCP":false,
 "MTE":false,
 "MT_EDT":false,
 "NORP":false,
 "PDIU":true,
 "PFDE":false,
 "PFDM":true,
```

```

 "QFQM":false,
 "QUOAC":true,
 "RDS":false,
 "RTTL":false,
 "RTTWP":false,
 "SSET":true,
 "TRACE":false,
 "TREU":true,
 "TRST":false,
 "TSCU":false,
 "UDBC":false,
 "UEIP":true,
 "VTIME":false
 }
}
},
 "TotalAssociations":1
}

admin@pod-cfgmgr-5-4-0-81-rel-4-2-0-5b85684cf6-kb6n4 09:19:17#

```

## Configure CLI support for PFCP Association

Configure the PFCP Peer. Log in to cnfmgr pod from fed-upf and add the following configuration:

```

pfcpc admin-state enabled

pfcpc association fqdn upf.affirmed.com

pfcpc association default-profile PFCPPROFILE1

pfcpc peer SMF-1

fqdn smf.affirmed.com

port 8805

ipv4-address 10.15.17.99

!

```

```

pfcf profile PFCPPROFILE1

pfcf-association-retry-interval 60000

pfcf-node-response-timeout 2000

pfcf-node-retransmit-count 3

pfcf-heartbeat-periodic-interval 60000

pfcf-heartbeat-down-threshold 5

pfcf-guard-time-interval 60000

pfcf-session-monitor-interval 300

!

```

**Note:** In the above example, PFCF-C External LB IP address is 10.15.17.99

### Clear all Associated Sessions

```

admin@pod-cfgmgr-1-0-0-417-main-int-dbg-5445858bf-2vgmk
06:10:07# request ppe v1 pfcf-association

clear cpNodeId <cpfNodeId>

[2020-04-27T14:23:38][oper-script] HERE actual_params_len:2

{

"SuccessResponse":"Delete Initiated"

}

```

### Spec reference

TS 3GPP 29.244 Rel 15

## Routing Management

This section describes Routing Management.

## BFD Management

BFD management on the UPF allows to configure the BFD profiles to manage the timers and other BFD attributes. Based on the configuration, it enables BFD protection automatically for all static routes configured in a Network Context as well as for all SH-BGP peering sessions that are generated by the SH-BGP configuration in that Network Context.

## Configuration

Use the following steps to configure BFD profiles and sessions:

- [CLI Support - Configuration of BFD Profiles](#)
- [Oper CLI Support - BFD Sessions](#)

## CLI Support - Configuration of BFD Profiles

**Note:** All static routes and BFD configurations are under **routing** node. BFD configuration is outside the network context and allows the user to configure transmit interval, receive interval, and detection multiplier for BFD. This is bound to network context. All BFD sessions in the context are created with this profile.

To configure the BFD profile, execute the following command:

```
bfd-session-profile <profile-name> <tx-interval> <rx-interval>
<multiplier>
```

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-94797bdb7-shx2w
04:58:17(config-routing)# bfd-session-profile NC-profile

Value for 'tx-interval' (<Transmit interval in milliseconds in
the range [100 .. 4294967].>): 1000
```

```
Value for 'rx-interval' (<Minimum receive interval in
milliseconds in the range [100 .. 4294967].>): 1000
```

```
Value for 'multiplier' (<The hello multiplier value in the
range [1 .. 255].>): 3
```

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-94797bdb7-shx2w
04:58:40(config-bfd-session-profile-NC-profile)# commit

Commit complete.
```

```
show running-config
```



```

upf service admin-state enabled

routing

bfd-session-profile NC-profile

tx-interval 1000

rx-interval 1000

multiplier 3

!

!

```

The following command provides configuration of BFD single-hop profile under network context:

```

routing

network-context nc1

bfd-sh-profile p1

!

```

**Note:** If *bfd-sh-profile* is not configured, BFD session is configured with default timers which is (tx=300, rx=300, multiplier=3).

## Oper CLI Support - BFD Sessions

To configure the BFD sessions using Oper CLI, execute the following command:

```

show routing v1 bfd-sessions network-context nc1

{
 "BFDSessions": [
 {
 "clients": "ribmgr",
 "destination-address": "10.164.4.33",
 "interface": "nc1-VLAN1504-VirtualFuncEthernet0-6-0-1",
 "ppeId": "ppe-1-3-0-411-main-int-dbg-6ccd5878b6-f82k9",
 "source-address": "10.164.4.34",
 "state": "Up"
 }
]
}

```

```

},
{
 "clients": "ribmgr",
 "destination-address": "2001:db8:0:1:10:164:4:33",
 "interface": "nc1-VLAN1504-VirtualFuncEthernet0-6-0-1",
 "ppeId": "ppe-1-3-0-411-main-int-dbg-6ccd5878b6-f82k9",
 "source-address": "2001:db8:0:1:10:164:4:34",
 "state": "Up"
}
]
}

show dataplane-agent v1 bfd-sessions
{
 "BFDSessions": [
 {
 "destination-address": "10.164.4.33",
 "detection-time": 900,
 "echo": "Disabled",
 "hop_type": "Single-Hop",
 "interface": "nc1-VLAN1504-VirtualFuncEthernet0-7-0-1",
 "local_diag": "No Diagnostic",
 "local_discriminator": "4155539298",
 "min_rx": 250,
 "min_tx": 250,
 "multiplier": 3,
 "network-context": "nc1",
 "remote_diag": "No Diagnostic",
 "remote_discriminator": "329",
 "remote_min_rx": 300,
 "remote_min_tx": 300,
 "remote_multiplier": 3,
 }
]
}

```

```

"source-address":"10.164.4.34",
"state":"Up",
"up-time":"00:06:54"
},
{
"destination-address":"2001:db8:0:1:10:164:4:33",
"detection-time":6000,
"echo":"Disabled",
"hop_type":"Single-Hop",
"interface":"ncl-VLAN1504-VirtualFuncEthernet0-7-0-1",
"local_diag":"No Diagnostic",
"local_discriminator":"273217928",
"min_rx":250,
"min_tx":250,
"multiplier":3,
"network-context":"ncl",
"remote_diag":"No Diagnostic",
"remote_discriminator":"500",
"remote_min_rx":2000,
"remote_min_tx":2000,
"remote_multiplier":3,
"source-address":"2001:db8:0:1:10:164:4:34",
"state":"Up",
"up-time":"00:06:26"
}
]
}
admin@xyz-10-163-65-6-07-02-05-52 05:50:15#

```

## Static Routing

Static Routing is configured in a specific Network Context. Optionally, you can enable BFD support for all the static routes configured.

### Configuration

- [CLI Support - Configuration of Static Routes](#)
- [CLI Support - Enabling BFD for Static Routes](#)
- [Oper CLI Support for Static Routes](#)

## Configuration of Static Routes

If UPF is either not directly connected to your gNodeB or the DUT on the N6 interface, configure the static routes to reach the destinations that the UPF needs to communicate with.

Following steps provide instructions on configuring static routes:

1. For creating a static route in a setup that does not use VLANs, execute the following command from the VPP container.

**For example:**

```
vppctl ip route add 10.192.18.60/24 via 10.10.5.2
VirtualFunctionEthernet0/6/0
```

2. For creating a static route in a setup that does use VLANs, execute the following command from the VPP container.

**For example:**

```
vppctl ip route add 10.192.18.60/24 via 10.10.5.2
VirtualFunctionEthernet0/6/0.100
```

## CLI Support - Configuration of Static Routes

***Note:** Static route configuration is under the network context.*

To configure the static routes, execute the following command:

```
ip-route <destination-address> <destination prefix-length>
<nexthop-address>
```

```
admin@pod-cfgmgr-1-0-0-200-main-int-dbg-94797bdb7-shx2w
05:02:58 (config-routing) # network-context NC
```

```

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-94797bdb7-shx2w
05:03:35 (config-network-context-NC) # ?

Possible completions:

bfd-all-static-enable Enable BFD sessions for all
IPv4/IPv6 static routes in this context

bfd-default-profile

bfd-static-routes-profile

ip-route Configuration of static IP routes

```

```

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-94797bdb7-shx2w
05:03:35 (config-network-context-NC) # ip-route 2.2.2.2 32
10.1.1.2

```

```

admin@pod-cfgmgr-1-0-0-200-main-int-dbg-94797bdb7-shx2w
05:08:48# show running-config

routing

network-context NC

ip-route 2.2.2.2 32 10.1.1.2

!

!

```

**Note:** Preferences and tags are not supported, though the CLI allows you to configure it.

## CLI Support - Enabling BFD for Static Routes

To enable BFD, configure keyword **bfd-all-static-enable** under network context. Optionally BFD profile can also be configured to set the desired BFD timers for the static BFD sessions.

**Note:** When BFD is enabled at the network context level, it is applicable to all static routes in the network context.

To configure the BFD for static routes, execute the following commands:

```

admin@xyz-10-163-65-6-07-02-05-52 05:34:06# show running-config
routing

routing

network-context ncl

```

```

bfd-all-static-enable

bfd-sh-profile p1

ip-route 111.1.1.1 32 10.164.4.33

!

ip-route 1111::1 128 2001:db8:0:1:10:164:4:33

!

!

bfd-session-profile p1

tx-interval 250

rx-interval 250

multiplier 3

!

!

admin@xyz-10-163-65-6-07-02-05-52 05:37:32#

```

**Note:** If *bfd-sh-profile* is not configured, BFD session is configured with default timers which is (tx=300, rx=300, multiplier=3).

## Oper CLI Support for Static Routes

To configure IPv4/IPv6 static routes, execute the following Oper CLI:

```

admin@xyz-10-163-65-6-07-02-05-52 11:29:30# show rib-mgr v1
ipv4-route network-context nc1

{
 "IpRoute": [
 {
 "destination-prefix": "10.164.4.32/29",
 "protocols": [
 {
 "Proto-name": "connected",
 "Selected": "yes",
 "nexthops": [

```

```

{
 "interface": "nc1-VLAN1504-VirtualFuncEthernet0-6-0-1",
 "state": "active"
}
]
}
]
},
{
 "destination-prefix": "111.1.1.1/32", <<< ipv4 static route
 "protocols": [
 {
 "Proto-name": "static",
 "Selected": "yes",
 "nexthops": [
 {
 "interface": "nc1-VLAN1504-VirtualFuncEthernet0-6-0-1",
 "next-hop": "10.164.4.33",
 "state": "active"
 }
]
 }
],
},
{
 "destination-prefix": "127.0.0.0/8",
 "protocols": [
 {
 "Proto-name": "connected",
 "Selected": "yes",
 "nexthops": [

```

```

{
 "interface": "lo-nc1",
 "state": "active"
}
]
}
]
}
]
}

admin@xyz-10-163-65-6-07-02-05-52 11:29:38#

admin@xyz-10-163-65-6-07-02-05-52 11:30:16# show rib-mgr vl
ipv6-route network-context nc1

{
 "IpRoute": [
 {
 "destination-prefix": "::1/128",
 "protocols": [
 {
 "Proto-name": "connected",
 "Selected": "yes",
 "nexthops": [
 {
 "interface": "lo-nc1",
 "state": "active"
 }
]
 }
]
 }
],
 {

```



```

"destination-prefix":"1111::1/128", <<< ipv6 static route
"protocols":[
{
 "Proto-name":"static",
 "Selected":"yes",
 "nexthops":[
 {
 "interface":"ncl-VLAN1504-VirtualFuncEthernet0-6-0-1",
 "next-hop":"2001:db8:0:1:10:164:4:33",
 "state":"active"
 }
]
}
],
{
 "destination-prefix":"2001:db8:0:1:10:164:4:30/125",
 "protocols":[
 {
 "Proto-name":"connected",
 "Selected":"yes",
 "nexthops":[
 {
 "interface":"ncl-VLAN1504-VirtualFuncEthernet0-6-0-1",
 "state":"active"
 }
]
 }
]
}
]

```

```
}
```

```
admin@xyz-10-163-65-6-07-02-05-52 11:30:21#
```

## Dynamic Routing

Dynamic Routing allows the template configuration of SH-BGP peers, where the actual peering configuration is generated dynamically for the interfaces on the PPE pod instances.

Dynamic Routing supports the following functions:

- SH-BGP with IPv4/IPv6 peering
- Template BGP peering configuration
- BFD sessions for generated BGP peers
- Loopbacks and UE Pools are configured on the UPF and advertised over Routing

***Note:** Dynamic Routing supports the UPF limits to SH-BGP and SH-BFD.*

## Configuration

Use the following steps to configure Dynamic Routing:

- [CLI Configuration for SH-BGP](#)
- [Oper CLI for BGP Summary](#)
- [Oper CLI for BGP Neighbors](#)
- [Oper CLI for BFD sessions for BGP Peers](#)
- [Oper CLI for BGP Routes](#)
- [CLI Support - Enabling BFD for Static Routes](#)
- [BGP Soft Clear and Hard Clear](#)

For route policies and configuring them, see [Route Policies](#).

## CLI Configuration for SH-BGP

The BGP template configuration allows the UPF to generate BGP peering configuration based on the PPE/Interfaces that are configured dynamically on the UPF.

To display the BGP running configurations, execute the following command:

```
admin@pod-cfgmgr-1-0-0-419-main-int-dbg-f467dc765-ht9cf
06:12:23# show running-config routing
routing
```

```

network-context nc1
 bfd-all-static-enable
 bfd-default-profile p1
 bgp
 local-as 100
 router-id 1.1.1.1

graceful-restart enable restart-time 90
>>>>> this is a default value, and can be modified

graceful-restart enable stalepath-time 360 >>>>>
this is a default value, and can be modified

 ipv4-ucast-af
 neighbor 10.164.4.33

 activate >>>> this is mandatory configuration
 nexthop-self
 capabilities graceful-restart

 !
 neighbor 10.164.4.41

 activate
 route-map in route-map-name v4_rmap_1
 !
 redistribute static
 !
 neighbor 10.164.4.33
 remote-as 200
 !
 neighbor 10.164.4.41
 remote-as 200
 !
 !
 !
 bfd-session-profile p1
 tx-interval 300
 rx-interval 300
 multiplier 3
 !
 !

```

### **For IPv6 BGP Neighbors:**

```

routing

network-context NC_N3

bgp

ipv6-ucast-af

```

```

neighbor 2001:10:10:10:1:1:1:254
activate
route-map out route-map-name v6_routeMap_1
!
neighbor 2001:10:10:11:1:1:1:254
activate
nexthop-self
capabilities graceful-restart
!
redistribute connected
redistribute static
!
!
!
!
```

### **For Enabling BFD Protection to BGP Neighbors**

```

routing
network-context NC_N6
bgp
local-as 1011
router-id 1.7.17.1
ipv4-ucast-af
neighbor 10.10.20.254
activate
!
redistribute connected
redistribute static
!
neighbor 10.10.20.254
remote-as 65000
fall-over-bfd
```

!  
!  
!

**Note:** Specify BFD session parameters in the **sh-profile** similar to static routes.  
PPE/interfaces with IP address **10.164.4.x** generates BGP peering configuration dynamically.  
You can configure the **sh-profile** . If it is not configured, then the default values are considered.

## Oper CLI for BGP Summary

To display the BGP summary, execute the following command:

```
admin@pod-cfgmgr-1-0-0-419-main-int-dbg-f467dc765-ht9cf
06:12:56# show bgp-mgr vl summary
network-context ncl

[2020-04-17T06:12:59][oper-script] HERE actual_params_len:2
{
 "Bgp-identifier":"1.1.1.1",
 "Community-entries":0,
 "Ebgp-multipath":64,
 "Ibgp-multipath":64,
 "Local-as":100,
 "Network-context":"ncl",
 "as-path-entries":1,
 "bgpSessions":[
 {
 "Neighbor":"10.164.4.33",
 "as-number":200,
 "message-received":6,
 "message-sent":4,
 "prefix-received":6,
 "state":"Established",
 "table-version":2,
 "up-time":"00:00:54"
 },
 {
 "Neighbor":"10.164.4.41",
 "as-number":200,
 "message-received":7,
 "message-sent":4,
 "prefix-received":6,
 "state":"Established",
 "table-version":2,
```

```

 "up-time":"00:00:54"
 }
]
}

```

## Oper CLI for BGP Neighbors

To display the BGP neighbors, execute the following command:

```

admin@pod-cfgmgr-1-0-0-419-main-int-dbg-f467dc765-ht9cf
06:13:11# show bgp-mgr v1
neighbor network-context ncl

[2020-04-17T06:13:18][oper-script] HERE actual_params_len:2
{
 "neighbors":[
 {
 "Local-as":200,
 "Neighbor":"10.164.4.33",
 "Remote-as":100,
 "Remote-routerId":"7.7.7.254",
 "State":"Established",
 "Up-time":"00:01:13",
 "address-family":[
 {
 "accepted-prefix":6,
 "address-family-name":"IPv4 Unicast",
 "announced-prefix":6,
 "bgp-table-version":2,
 "community-attribute":"(both)",
 "neighbor-version":2
 }
],
 "capabilities-4-Octet-ASN":"received",
 "capabilities-address-family-ipv4-unicast":"advertised and received",
 "capabilities-route-refresh":"advertised and received (old and new)",
 "connections":"Established 1, Dropped 0",
 "local-host":"10.164.4.34",
 "local-port":50708,
 "messages-keepalives":"Received 3, Sent 3",
 "messages-notifications":"Received 0, Sent 0",
 "messages-received":7,
 "messages-route-refresh":"Received 0, Sent 0",
 "messages-sent":5,

```

```

 "nexthop-ifindex":1,
 "remote-host":"10.164.4.33",
 "remote-port":179,
 "timers":"hold time is 90 seconds, keepalive
interval is 30 seconds,
advertisement-interval is 30 seconds"
 },
 {
 "Local-as":200,
 "Neighbor":"10.164.4.41",
 "Remote-as":100,
 "Remote-routerId":"7.7.7.254",
 "State":"Established",
 "Up-time":"00:01:13",
 "address-family":[
 {
 "accepted-prefix":6,
 "address-family-name":"IPv4 Unicast",
 "announced-prefix":6,
 "bgp-table-version":2,
 "community-attribute":"(both)",
 "neighbor-version":2
 }
],
 "capabilities-4-Octet-ASN":"received",
 "capabilities-address-family-ipv4-
unicast":"advertised and received",
 "capabilities-route-refresh":"advertised and
received (old and new)",
 "connections":"Established 1, Dropped 0",
 "local-host":"10.164.4.42",
 "local-port":34822,
 "messages-keepalives":"Received 3, Sent 3",
 "messages-notifications":"Received 0, Sent 0",
 "messages-received":7,
 "messages-route-refresh":"Received 0, Sent 0",
 "messages-sent":5,
 "nexthop-ifindex":2,
 "remote-host":"10.164.4.41",
 "remote-port":179,
 "timers":"hold time is 90 seconds, keepalive
interval is 30 seconds,
advertisement-interval is 30 seconds"
 }
]
}

```



## Oper CLI for BFD sessions for BGP Peers

To display BFD sessions for BGP Peers, execute the following command:

```
admin@xyz-10-163-65-6-07-02-05-52 05:48:37# show routing v1
bfd-sessions network-context nc1
```

```
{
 "BFDSessions": [
 {
 "clients": "bgpMgr",
 "destination-address": "10.164.4.33",
 "interface": "nc1-VLAN1504-VirtualFuncEthernet0-7-0-1",
 "ppeId": "ppe-1-3-0-411-main-int-dbg-7b64cb854c-q5tfp",
 "source-address": "10.164.4.34",
 "state": "Up"
 },
 {
 "clients": "bgpMgr",
 "destination-address": "2001:db8:0:1:10:164:4:33",
 "interface": "nc1-VLAN1504-VirtualFuncEthernet0-7-0-1",
 "ppeId": "ppe-1-3-0-411-main-int-dbg-7b64cb854c-q5tfp",
 "source-address": "2001:db8:0:1:10:164:4:34",
 "state": "Up"
 }
]
}
```

```
admin@xyz-10-163-65-6-07-02-05-52 05:49:03#
```

```
admin@xyz-10-163-65-6-07-02-05-52 05:50:10# show dataplane-
agent v1 bfd
```

```
{
 "BFDSessions": [
 {
```

```

"destination-address":"10.164.4.33",
"detection-time":900,
"echo":"Disabled",
"hop_type":"Single-Hop",
"interface":"nc1-VLAN1504-VirtualFuncEthernet0-7-0-1",
"local_diag":"No Diagnostic",
"local_discriminator":"4155539298",
"min_rx":250,
"min_tx":250,
"multiplier":3,
"network-context":"nc1",
"remote_diag":"No Diagnostic",
"remote_discriminator":"329",
"remote_min_rx":300,
"remote_min_tx":300,
"remote_multiplier":3,
"source-address":"10.164.4.34",
"state":"Up",
"up-time":"00:06:54"
},
{
"destination-address":"2001:db8:0:1:10:164:4:33",
"detection-time":6000,
"echo":"Disabled",
"hop_type":"Single-Hop",
"interface":"nc1-VLAN1504-VirtualFuncEthernet0-7-0-1",
"local_diag":"No Diagnostic",
"local_discriminator":"273217928",
"min_rx":250,
"min_tx":250,
"multiplier":3,

```

```

"network-context":"nc1",
"remote_diag":"No Diagnostic",
"remote_discriminator":"500",
"remote_min_rx":2000,
"remote_min_tx":2000,
"remote_multiplier":3,
"source-address":"2001:db8:0:1:10:164:4:34",
"state":"Up",
"up-time":"00:06:26"
}
]
}

admin@xyz-10-163-65-6-07-02-05-52 05:50:15#

```

## Oper CLI for BGP Routes

To display the BGP routes in the below mentioned address family, execute specific command as follows:

```

show bgp-mgr v1 ipv4-ucast-af network-context NC-name
detailed

show bgp-mgr v1 ipv6-ucast-af network-context NC-name
detailed

show bgp-mgr v1 vpv4-ucast-af network-context NC-name
detailed

show bgp-mgr v1 vpv6-ucast-af network-context NC-name
detailed

```

*Note: Here, detailed is an optional parameter.*

## Sample Output

Following is the sample output for BGP routes:

```

show bgp-mgr v1 ipv4-ucast-af network-context N6
{

```

```

"Source-PPEID":"ppe-4-1-0-1205-main-int-sym-6884dbf848-55w2h",
"bgpRoutes":[
{
"destination-prefix":"1.2.3.4/32",
"next-hops":[
{
"best":"yes",
"nexthop":"127.0.0.2",
"origin":"?",
"path":"",
"valid":"yes",
"weight":32768
}
]
},
{
"destination-prefix":"10.10.71.0/24",
"next-hops":[
{
"best":"yes",
"nexthop":"10.10.73.2",
"origin":"i",
"path":"200",
"valid":"yes",
"weight":0
}
]
},
{
"destination-prefix":"10.10.73.0/24",
"next-hops":[

```

```

{
 "best": "yes",
 "nexthop": "0.0.0.0",
 "origin": "?",
 "path": "",
 "valid": "yes",
 "weight": 32768
},
{
 "best": "no",
 "nexthop": "10.10.73.2",
 "origin": "i",
 "path": "200",
 "valid": "yes",
 "weight": 0
}
],
{
 "destination-prefix": "20.20.1.0/24",
 "next-hops": [
 {
 "best": "yes",
 "nexthop": "10.10.74.2",
 "origin": "?",
 "path": "",
 "valid": "yes",
 "weight": 32768
 }
]
},

```

```
{
 "destination-prefix":"99.1.1.1/32",
 "next-hops":[
 {
 "best":"yes",
 "nexthop":"10.10.73.2",
 "origin":"i",
 "path":"200",
 "valid":"yes",
 "weight":0
 }
]
}
```

## Optional Detailed Sample Output

```
show bgp-mgr v1 ipv4-ucast-af network-context N6 detailed

{
 "Source-PPEID":"ppe-4-1-0-1205-main-int-sym-6884dbf848-55w2h",
 "bgpRoutes":[
 {
 "destination-prefix":"1.2.3.4/32",
 "next-hops":[
 {
 "best":"yes",
 "extensive-route-info":[
 {}
],
 "nexthop":"127.0.0.2",
 "origin":"?",

```

```

"path": "",
"valid": "yes",
"weight": 32768
}
],
{
"destination-prefix": "10.10.71.0/24",
"next-hops": [
{
"best": "yes",
"extensive-route-info": [
{
"ext-community": [
"RT: 3001:1 "
]
}
],
"nexthop": "10.10.73.2",
"origin": "i",
"path": "200",
"valid": "yes",
"weight": 0
}
],
{
"destination-prefix": "10.10.73.0/24",
"next-hops": [
{
"best": "yes",

```

```

"extensive-route-info":[
 {}
],
"nexthop":"0.0.0.0",
"origin":"?",
"path":"",
"valid":"yes",
"weight":32768
},
{
 "best":"no",
 "extensive-route-info":[
 {
 "ext-community":[
 "RT: 3001:1 "
]
 }
],
 "nexthop":"10.10.73.2",
 "origin":"i",
 "path":"200",
 "valid":"yes",
 "weight":0
}
],
{
 "destination-prefix":"20.20.1.0/24",
 "next-hops":[
 {
 "best":"yes",

```



```

"extensive-route-info":[
 {}
],
"nexthop":"10.10.74.2",
"origin":"?",
"path":"",
"valid":"yes",
"weight":32768
}
],
{
 "destination-prefix":"99.1.1.1/32",
 "next-hops":[
 {
 "best":"yes",
 "extensive-route-info":[
 {
 "ext-community":[
 "RT: 3001:1 "
]
 }
],
 "nexthop":"10.10.73.2",
 "origin":"i",
 "path":"200",
 "valid":"yes",
 "weight":0
 }
]
}

```

```
]
}
```

## CLI Support - Enabling BFD for BGP Peers

To enable BFD for BGP peers, execute the following command:

```
admin@xyz-10-163-65-6-07-02-05-52 05:47:47# show running config

routing
routing
network-context ncl
bfd-sh-profile p1
bgp
local-as 100
router-id 1.1.1.1
ipv4-ucast-af
neighbor 10.164.4.33
activate
!
!ipv6-ucast-af
neighbor 2001:db8:0:1:10:164:4:33
activate
!
!
neighbor 10.164.4.33
remote-as 200
fall-over-bfd
!
neighbor 2001:db8:0:1:10:164:4:33
remote-as 200
```

```
fall-over-bfd
!
!
!
bfd-session-profile p1
tx-interval 250
rx-interval 250
multiplier 3
!
!
admin@xyz-10-163-65-6-07-02-05-52 05:48:22#
```

**Note:** If *bfd-sh-profile* is not configured, BFD session is configured with default timers which is (tx=300, rx=300, multiplier=3).

# BGP Soft Clear and Hard Clear

## Hard Clear

The hard clear command resets the TCP sessions of a particular network-context.

```
request bgp-mgr v1 clear all hard network-context <nc_name>
<<<<<<<<<<<< clears and brings up bgp session
```

## Soft Clear

The soft clear command refreshes the routes without resetting the TCP sessions in a particular network-context.

```
request bgp-mgr v1 clear all soft network-context <nc_name>
<<<<<<<<<<<< used to send the route update
```

# Routing Policies

Routing policies control the decision of importing/exporting the routes into the routing table.

Following are the functions of Routing Policies:

- Matches **Rules** of the routing policies will help to control the prefixes into the routing table, both inbound and outbound.
- Sets **Rules** of the routing policies will help the routing protocols to change/set the route attributes.
- Includes **ip-prefix-lists**, **access-lists**, **aspath-lists**, **route-maps** and so on.

***Note:** For a given protocol/session, one or more policies can be applied to control the route updates while advertising or receiving.*

## Configuration

The following sections provide configuration examples for Routing Policies:

### IPv4 Prefix list

```
routing
policy
prefix-list v4list1
1 permit 10.10.12.0 24 0 0
2 deny 10.10.13.0 24 0 0
!
!
!
```

### IPv6 Prefix List

```
routing
policy
prefix-list-ipv6 v6list1
1 deny 2001:10:10:12:: 64 0 0
2 permit 3001:10:10:12:: 64 0 0
!
!
```

!

## Route-Map

```
routing
policy
route-map v6_routeMap_1 deny 1
match-ipv6-address-prefix-list v6list1
!
!
!
routing
policy
route-map v4_rmap_1 deny 1
match-ip-address-prefix-list v4list2
!
!
!
```

| CLI Command                                                                       | Description                                                         |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------|
| (no) routing policy prefix-list <NAME>                                            | Configure an ipv4 prefix list                                       |
| The below attributes are configured under routing policy prefix-list <NAME>       |                                                                     |
| (no) <SEQNO> permit address <ADDR> prefix-length <PFXLEN> {ge <GE>  } {le <LE>  } | Configure the ip of the prefix with prefix-length with permit rule. |
| (no) <SEQNO> deny address <ADDR> prefix-length <PFXLEN> {ge <GE>  } {le <LE>  }   | Configure the ip of the prefix with prefix-length with deny rule.   |
| (no) routing policy prefix-list <NAME> description <TEXT>                         | To set the policy name description                                  |
| no) routing policy prefix-list-ipv6 <NAME>                                        | Configure an ipv6 prefix list                                       |
| The below attributes are configured under routing policy prefix-list-ipv6 <NAME>  |                                                                     |
| (no) <SEQNO> permit address <ADDR> prefix-length <PFXLEN> {ge <GE>  } {le <LE>  } | Configure the ip of the prefix with prefix-length with permit rule. |
| (no) <SEQNO> deny address <ADDR> prefix-length <PFXLEN> {ge <GE>  } {le <LE>  }   | Configure the ip of the prefix with prefix-length with deny rule.   |

| CLI Command                                                                                              | Description                                                                       |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| (no) routing policy prefix-list-ipv6 <NAME> description <TEXT>                                           | To set the policy name description                                                |
| (no) routing policy route-map <NAME> { permit   deny } <SEQNO>                                           | Create a route-map with permit or deny rule, and with the unique sequence number. |
| The below attributes shall be configured under routing policy route-map <NAME> { permit   deny } <SEQNO> |                                                                                   |
| (no) match-ip-address-prefix-list <NAME>                                                                 | BGP prefix list match by name                                                     |
| (no) match-ipv6-address-prefix-list <NAME>                                                               | BGP IPV6 prefix list match by name                                                |

## Route Policies

Routing policies are used to control the decision of importing or exporting the routes into the routing table. Routing policies include ip-prefix-lists, community lists, and route-maps. Route-map can be applied per neighbor per address-family.

The attributes of route-maps are:

- Match Rules of the routing policies will help to control the prefixes into the routing table, both inbound and outbound.
- Set Rules of the routing policies will help the routing protocols to change or set the route attributes.

## Configuration

The following are the commands to configure different types of route policies:

**This section describes configuration of Prefix-list and route-maps**

```

routing
policy
prefix-list v4Plist_1
1 permit address 11.11.0.0 prefix-length 16 le 0 ge 0
prefix-list-ipv6 v6Plist_1
1 permit address 2001:10:11:11:: prefix-length 64 le 0 ge 0
route-map v4Rmap_1
permit 1
match-ip-address-prefix-list v4Plist_1

```

```

!
!

route-map v6Rmap_1

permit 1

match-ipv6-address-prefix-list v6Plist_1

admin# config

(config)# routing network-context N3 bgp ipv4-ucast-
af neighbor 10.10.20.254 route-map out route-map-name
v4Rmap_1

(config-neighbor-10.10.20.254)# commit

Commit complete. (config-neighbor-10.10.20.254)# end

config

(config)# routing network-context N3 bgp ipv6-ucast-
af neighbor 2001:10:10:20:1:1:1:254 route-map in
route-map-name v6Rmap_1

(config-neighbor-2001:10:10:20:1:1:1:254)# commit

Commit complete.

(config-neighbor-2001:10:10:20:1:1:1:254)# end

admin#

```

## Sample Output

```

show running-config routing network-context N3 bgp
ipv4-ucast-af neighbor 10.10.20.254

routing

network-context N3

bgp

ipv4-ucast-af

neighbor 10.10.20.254

activate

route-map out route-map-name v4Rmap_1

```



```

!
show running-config routing network-context N3 bgp
ipv6-ucast-af neighbor 2001:10:10:20:1:1:1:254

routing

network-context N3

bgp

ipv6-ucast-af

neighbor 2001:10:10:20:1:1:1:254

activate

route-map in route-map-name v6Rmap_1

!

12:56:38#

```

**This section describes configuration of set set-ipv6-next-hop/set-ip-next-hop**

```

routing

policy route-map setVpnV6Nh

permit 10

set-ipv6-next-hop 2001:10:10:10:1:1:1:254

config

Entering configuration mode terminal

(config)# routing policy route-map setVpnV4Nh permit
10 set-ip-next-hop 10.10.10.22

(config-route-map-permit/10)# commit

Commit complete.

(config-route-map-permit/10)# end

```

## Sample Output

```

show running-config routing network-context N6 bgp
vpnv6-ucast-af

routing

network-context N6

bgp

vpnv6-ucast-af

```

```

neighbor 10.10.10.254
activate
nexthop-self
capabilities graceful-restart
send-community both
route-map in route-map-name setVpnV6Nh
!
neighbor 10.10.11.254
activate
nexthop-self
capabilities graceful-restart
send-community both
route-map in route-map-name setVpnV6Nh_1
!
config
Entering configuration mode terminal

(config)# routing network-context N6 bgp vpnv4-ucast-
af neighbor 10.10.10.254 route-map out route-map-name
Possible completions:
out_setVpnV6Nh out_setVpnV6Nh_1 setVpnV4Nh
setVpnV6Nh setVpnV6Nh_1 v4Rmap_1 v6Rmap_1

(config)# routing network-context N6 bgp vpnv4-ucast-
af neighbor 10.10.10.254 route-map out route-map-name
setVpnV4Nh

(config-neighbor-10.10.10.254)# commit

Commit complete.

show running-config routing network-context N6 bgp
vpnv4-ucast-af neighbor 10.10.10.254

routing

network-context N6

bgp

vpnv4-ucast-af

```

```

neighbor 10.10.10.254

activate

nexthop-self

capabilities graceful-restart

send-community both

route-map out route-map-name setVpnV4Nh

!
```

**This section describes the configuration of *comm-list/comm-list-ext/comm-list-word/comm-list-word-ext***

```

config

routing policy comm-list 10

(config-comm-list-10)# 100:10 permit

(config-comm-list-10)#

(config-comm-list-10)# comm-list-ext 110

(config-comm-list-ext-110)# 4000:100 permit

(config-comm-list-ext-110)#

(config-comm-list-ext-110)# comm-list-word
stdCommList1

(config-comm-list-word-stdCommList1)# no-advertise
permit

(config-comm-list-word-stdCommList1)#

(config-comm-list-word-stdCommList1)# comm-list-word-
ext extCommList1

(config-comm-list-word-ext-extCommList1)# 3000:100
permit

(config-comm-list-word-ext-extCommList1)#

(config-comm-list-word-ext-extCommList1)# route-map
rMapComm_1 permit 1

(config-route-map-rMapComm_1/permit/1)# match-comm-
list 10

(config-route-map-rMapComm_1/permit/1)# end

config
```

```

Entering configuration mode terminal

(config)# routing network-context N6 bgp vpnv4-ucast-
af neighbor 10.10.10.254 route-map in route-map-name
rMapComm_1

(config-neighbor-10.10.10.254)# commit

Commit complete.

(config-neighbor-10.10.10.254)# end

```

This section describes the route-map configuration with different match rules

```

config

Entering configuration mode terminal

(config)# routing policy route-map matchV4NHMap
permit 10 match-ip-next-hop 2.2.2.2

(config-route-map-permit/10)# exit

(config-route-map-matchV4NHMap)# exit

(config-policy)# route-map matchV6NHMap permit 20
match-ipv6-next-hop 2001:10:10:10:1:1:1:33

(config-route-map-permit/20)# exit

(config-route-map-matchV6NHMap)# exit

(config-policy)# route-map matchTagValueMap permit 30
match-tag 30

(config-route-map-permit/30)# commit

Commit complete.

01:12:05 (config-route-map-permit/30)# end

```

## Sample Output

```

show running-config routing policy route-map

routing

policy

route-map matchTagValueMap

permit 30

match-tag 30

!

```

```

route-map matchV4NHMap
permit 10
match-ip-next-hop 2.2.2.2
!
route-map matchV6NHMap
permit 20
match-ipv6-next-hop 2001:10:10:10:1:1:1:33
!

```

**This section describes the configuration of route-map with different set rules**

```
config
```

Entering configuration mode terminal

```
(config)# routing policy route-map setCommunityMap
permit 100 set-community ?
```

Possible completions:

```

additive Add to the existing
community
community-number
community-number-alt-format
well-known-communities

```

```
(config)# routing policy route-map setCommunityMap
permit 100 set-community community-number ?
```

Possible completions:<1-4294967295> community  
number

```
(config)# routing policy route-map setCommunityMap
permit 100 set-community community-number 100
```

```
(config-route-map-permit/100)# commit
```

Commit complete.

```
(config-route-map-permit/100)# exit
```

```
(config-route-map-setCommunityMap)# exit
```

```
(config-policy)# route-map setExtCommMap permit 110
set-ext-community rt ?
```

```

Possible completions:<Regular Expression for route
target AA:NN>

(config-policy)# route-map setExtCommMap permit 110
set-ext-community rt 1000:100

(config-route-map-permit/110)# commit

Commit complete.

(config-route-map-permit/110)# exit

(config-route-map-setExtCommMap)# exit

(config-policy)# route-map setLocalPrefMap permit 200
set-local-preference ?

Description: Set BGP local preference path attribute

Possible completions:<unsignedInt>

(config-policy)# route-map setLocalPrefMap permit 200
set-local-preference 200

(config-route-map-permit/200)# exit

(config-route-map-setLocalPrefMap)# exit

(config-policy)# route-map setTagMap permit 210 set-
tag 20

(config-route-map-permit/210)# commit

Commit complete.

(config-route-map-permit/210)# end

```

## Sample Output

```

show running-config routing policy route-map

routing

policy

route-map setCommunityMap

permit 100

set-community community-number 100

!

route-map setExtCommMap

```

```
permit 110
set-ext-community rt 1000:100
!
route-map setLocalPrefMap
permit 200
set-local-preference 200
!
route-map setTagMap
permit 210
set-tag 20
!
```

# NRF Registration

The following Oper CLI commands displays the registration status and the configuration on the UPF that is part of the NRF registration.

## Sample running-configuration for NRF Registration

```
upf service admin-state enabled

upf service upf-fqdn upf.cluster41.com

upf service smf-serving-area AREALIST

upf service home-plmn-list PLMNLIST1

upf service allowed-nssai NSSAILIST

upf service network-context NC_N3
!

upf service network-context NC_N6
!

upf service-construct network-context NC_N3
 type N3
!

upf service-construct network-context NC_N6
 type N6
 ip-pool IP-POOL-LS21
 type UE-IP-Pool
 range-start 5.0.0.1
 range-end 5.15.255.254
 route-summary-prefix-length 12
 !
!

upf service-construct serving-area MYAREA
!

upf service-construct serving-area-list AREALIST
 serving-area MYAREA
!
```



```

!
upf service-construct s-nssai SNSSAI1
 slice-service-type 4
 dnn-list DNNLIST1
!
upf service-construct s-nssai SNSSAI2
 slice-service-type 5
 dnn-list DNNLIST2
!
upf service-construct nssai NSSAILIST
 s-nssai SNSSAI1
 !
 s-nssai SNSSAI2
 !
!
upf service-construct dnai upf.cluster41
!
upf service-construct dnai-list DNAI1
 dnai upf.cluster41
 !
!
upf service-construct dnn internet.dn41.net
 dnai-list DNAI1
!
upf service-construct dnn internet.test9.net
 dnai-list DNAI1
!
upf service-construct dnn-list DNNLIST1
 dnn internet.dn41.net
 !
!

```

```

upf service-construct dnn-list DNNLIST2
 dnn internet.test9.net
 !
!
upf service-construct plmn 310150
 nssai NSSAILIST
 !
upf service-construct plmn 310410
 nssai NSSAILIST
 !
upf service-construct plmn-list PLMNLIST1
 plmn 310150
 !
 plmn 310410
 !
!
.....
nrf-agent nrf-profile CL21-NRF
 plmn-info 310150
 !
 services nnrf-nfm
 version api-version-uri v1
 ipv4-address 10.72.10.21
 port 31780
 !
!

```

## Oper CLI to show nf-status

```

show upfmonitor nf-status
{
 "NfInstanceId":"7c73b931-9ceb-4aaf-9e5e-5f090ac04c79",

```

```

 "active":true,
 "applicationStatus":"READY",
 "deregistrationTime":"Thu, 23 Apr 2020 10:28:30 GMT",
 "monitorCurrentState":"Registered",
 "monitorPreviousState":"Deregistered",
 "nfPersonalityUpdateStatus":false,
 "overallContinuityCriticalServicesStatus":true,
 "overallReadyCriticalServicesStatus":true,
 "recoveryTime":"Fri, 17 Apr 2020 10:05:59 GMT",
 "restartCount":1
 }

```

## Oper CLI to show nf-profile

```
show upfmonitor nf-profile
```

```

{
 "allowedNssais":[
 {
 "sd":"bbbbbb",
 "sst":2
 },
 {
 "sd":"cccccc",
 "sst":1
 }
],
 "capacity":225,
 "fqdn":"affirmedUpf.com",
 "ipv4Addresses":[
 {
 "value":"10.163.206.162"
 }
]
}

```

```

],
"load":0,
"nfInstanceId":"7c73b931-9ceb-4aaf-9e5e-5f090ac04c79",
"nfServices":[
 {
 "apiPrefix":"","
 "capacity":0,
 "fqdn":"affirmedUpf.com",
 "interPlmnFqdn":"","
 "ipEndpoints":[
 {
 "ipv4Address":{
 "value":"10.163.206.162"
 },
 "ipv6Address":{
 "value":""
 },
 "port":8805,
 "transport":"UDP"
 }
],
 "load":0,
 "nfServiceStatus":"REGISTERED",
 "priority":0,
 "recoveryTime":"1601-01-01T00:00:00Z",
 "scheme":"http",
 "serviceInstanceId":"7c73b931-9ceb-4aaf-9e5e-5f090ac04c79-nupf-ppe",
 "serviceName":"nupf-ppe",
 "supportedFeatures":"","
 "versions":[

```

```

 {
 "apiFullVersion":"1.0.0.alpha-1",
 "apiVersionInUri":"v1",
 "expiry":"1601-01-01T00:00:00Z"
 }
]
}

],
"nfStatus":"REGISTERED",
"nfType":{
 "value":"UPF"
},
"plmnList":[
 {
 "mcc":"222",
 "mnc":"222"
 },
 {
 "mcc":"333",
 "mnc":"333"
 }
],
"priority":500,
"recoveryTime":"2020-04-24T14:57:38.857525Z",
"sNssais":[
 {
 "sd":"bbbbbb",
 "sst":2
 },
 {
 "sd":"cccccc",

```

```

 "sst":1
 }
],
"upfInfo":{
 "iwkEpsInd":true,
 "pduSessionTypes":[
 {
 "value":"IPv4"
 },
 {
 "value":"IPv6"
 },
 {
 "value":"IPv4V6"
 }
],
 "sNssaiUpfInfoList":[
 {
 "dnnUpfInfoList":[
 {
 "dnaiList":[
 {
 "value":"DNAI3"
 },
 {
 "value":"DNAI4"
 }
],
 "dnn":"DNN2",
 "pduSessionTypes":[
 {

```

```

 "value": "IPV4"
 },
 {
 "value": "IPV6"
 },
 {
 "value": "IPV4V6"
 }
]
}
],
"sNssai": {
 "sd": "bbbbbb",
 "sst": 2
}
},
{
 "dnnUpfInfoList": [
 {
 "dnaiList": [
 {
 "value": "DNAI5"
 },
 {
 "value": "DNAI6"
 }
],
 "dnn": "DNN3",
 "pduSessionTypes": [
 {
 "value": "IPV4"
 }
]
 }
]
}

```

```

 },
 {
 "value": "IPV6"
 },
 {
 "value": "IPV4V6"
 }
]
}

],
"sNssai": {
 "sd": "cccccc",
 "sst": 1
}
}

],
"smfServingArea": [
 "SA3",
 "SA4"
]
}
}

```

## Oper CLI to show provisioned smf-serving-area

```
#show upf v1 provisioned smf-serving-area
```

```

{
 "SmfServingArea": [
 {
 "ServingArea": [
 {
 "name": "SA3"
 }
]
 }
]
}

```



```

 },
 {
 "name": "SA4"
 }
],
 "name": "SAL2"
}
]
}

```

## Oper CLI to show provisioned PLMN

```

show upf v1 provisioned plmn
{
 "PlmnEntry": [
 {
 "name": "111111",
 "nssaiList": [
 {
 "Nssai": [
 {
 "dnn-list": "DNNL1",
 "name": "SNSSAI1",
 "slice-differentiator": "aabbcc",
 "slice-service-type": 3
 }
],
 "name": "NSL1"
 }
]
 },
 {

```

```

"name": "222222",
"nssaiList": [
{
"Nssai": [
{
"dnn-list": "DNNL2",
"name": "SNSSAI2",
"slice-differentiator": "bbbbbb",
"slice-service-type": 2
}
],
"name": "NSL2"
}
],
{
"name": "333333",
"nssaiList": [
{
"Nssai": [
{
"dnn-list": "DNNL3",
"name": "SNSSAI3",
"slice-differentiator": "cccccc",
"slice-service-type": 1
}
],
"name": "NSL3"
}
]
}

```

```
]
}
```

## Oper CLI to show provisioned home-plmn-list

```
show upf v1 provisioned home-plmn-list
```

```
{
 "Home-plmn":[
 {
 "Plmn":[
 {
 "name":"222222"
 },
 {
 "name":"333333"
 }
],
 "name":"PL2"
 }
]
}
```

## Oper CLI to show provisioned dnn-list

```
show upf v1 provisioned dnn-list
```

```
{
 "dnnList":[
 {
 "dnn":[
 {
 "dnai-list":[
 {
 "dnai":[
```

```

 {
 "name": "DNAI1"
 },
 {
 "name": "DNAI2"
 }
],
 "name": "DNAI1"
},
],
 "name": "DNN1"
},
],
 "name": "DNNL1"
},
{
 "dnn": [
 {
 "dnai-list": [
 {
 "dnai": [
 {
 "name": "DNAI3"
 },
 {
 "name": "DNAI4"
 }
],
 "name": "DNAI2"
 }
],

```

```

 "name": "DNN2"
 },
 {
 "name": "DNNL2"
 },
 {
 "dnn": [
 {
 "dnai-list": [
 {
 "dnai": [
 {
 "name": "DNAI5"
 },
 {
 "name": "DNAI6"
 }
]
 },
 {
 "name": "DNAI3"
 }
]
 },
 {
 "name": "DNN3"
 }
],
 "name": "DNNL3"
 }
]
}

```

## Oper CLI to show provisioned allowed-visited-plmn-list

```
show upf v1 provisioned allowed-visited-plmn-list
```

```

{
 "Allowed-plmn": [
 {
 "Plmn": [
 {
 "name": "111111"
 }
],
 "name": "PL1"
 }
]
}

```

## Oper CLI to show provisioned interface

**# show upf v1 provisioned interface**

```

{
 "Network-Contexts": [
 {
 "Interfaces": [
 {
 "if-id": 7,
 "ipv4-addresses": [
 {
 "ipv4": "10.164.4.34",
 "prefix-length": 29
 }
],
 "l3-mtu": 1500,
 "network-context": "ncl",

```

```

 "vlan-id":1504
 },
 {
 "if-id":8,
 "ipv4-addresses":[
 {
 "ipv4":"10.164.4.42",
 "prefix-length":29
 }
],
 "l3-mtu":1500,
 "network-context":"ncl",
 "vlan-id":1505
 },
 {
 "if-id":9,
 "ipv4-addresses":[
 {
 "ipv4":"10.164.4.202",
 "prefix-length":30
 }
],
 "l3-mtu":1500,
 "network-context":"ncl",
 "vlan-id":1571
 },
 {
 "if-id":10,
 "ipv4-addresses":[
 {
 "ipv4":"10.164.4.206",

```

```

 "prefix-length":30
 }
],
 "l3-mtu":1500,
 "network-context":"nc1",
 "vlan-id":1580
}
],
 "index":44,
 "name":"nc1"
}
]
}

```

## Oper CLI to show provisioned UE ip-pool

**# show upf vl provisioned ip-pool**

```

{
 "Network-Contexts":[
 {
 "IP Pools":[
 {
 "end-ip":"10.10.10.100",
 "name":"pool1",
 "route-summary-prefix-length":0,
 "start-ip":"10.10.10.1",
 "type":"UE-IP-Pool"
 },
 {
 "end-ip":"1000:1010::",
 "name":"pool2",
 "route-summary-prefix-length":0,

```



```

 "start-ip":"1000:1001::",
 "type":"UE-IP-Pool"
 },
 {
 "end-ip":"40.40.40.1",
 "name":"pool3",
 "route-summary-prefix-length":0,
 "start-ip":"30.30.30.1",
 "type":"UE-IP-Pool"
 }
],
"name":"nc1"
},
{
 "name":"nc2"
}
]
}

```

## Support for TAI List Configuration and NRF Registration

This section provides following details:

- Support for **TAI** and **TAI List** configuration under **upf service-construct**
- Support for **upfInfo** and **upf-info-list** – Configurable under **upf service-construct**
- Configure **primary-upf-info** and **upf-info-list** to be used for NRF registration under **upf-service**

### TAI and TAI List Configuration

The following command shows TAI configuration:

```

admin@-# show running-config upf service-construct tai
upf service-construct tai TAI1
plmn-id 456789
tac 123ffa

```

```

!
upf service-construct tai TAI2
plmn-id 23456
tac ddffaa
!
upf service-construct TAI3
plmn-id 789124
tac aabbcc
!

```

The following command shows TAI List configuration:

```

admin@# show running-config upf service-construct tai-list
upf service-construct tai-list TAI-LIST1
tai TAI1
!
!
upf service-construct tai-list TAI-LIST2
tai TAI2
!
!
upf service-construct tai-list TAI-LIST3
tai TAI1
!
tai TAI3
!
!

```

## TAC Range Configuration

```

admin@-# show running-config upf service-construct tac-range
upf service-construct tac-range TAC-RANGE1
start 123ffa
end 123ffd
!

```

```

upf service-construct tac-range TAC-RANGE2

start ddffaa
end ddffba
!

upf service-construct tac-range TAC-RANGE3

start aabbcc
end aabbcf
!

```

### **TAC Range List Configuration**

```

admin@# show running-config upf service-construct tac-range-
list

upf service-construct tac-range-list TAC-LIST1
tac-range TAC-RANGE1
!
!

upf service-construct tac-range-list TAC-LIST2
tac-range TAC-RANGE2
!
!

upf service-construct tac-range-list TAC-LIST3
tac-range TAC-RANGE1
!
tac-range TAC-RANGE3
!
!

```

### **TAI Range Configuration**

```

admin@-# show running-config upf service-construct tai-range

upf service-construct tai-range TAI-RANGE1

plmn-id 123456

tac-range-list TAC-LIST1
!

```

```

upf service-construct tac-range TAI-RANGE2

plmn-id 789123

tac-range-list TAC-LIST3

!

```

## TAI Range List Configuration

```

admin@-# show running-config upf service-construct tai-range-
list

upf service-construct tai-range-list TAI-RANGE-LIST-1

tai-range TAI-RANGE1

!

tai-range TAI-RANGE2

!

```

## Upf-info and Upf-info-list Configuration

Execute the following command to display upf-info configuration:

```

admin@xyz-10-163-66-96-04-16-15-26 06:12:48# show running-
config upf service-construct upf-info

upf service-construct upf-info FOO

admin-state enabled

priority 10

iwk-eps-ind true

ueip-addr-ind true

smf-serving-area SA-List1

snssai-upfinfolist snssai-List1

tai-list TAI-LIST1

tai-range-list TAI-RANGE-LIST1

!

upf service-construct upf-info BAR

admin-state enabled

priority 20

iwk-eps-ind true

ueip-addr-ind true

```

```
smf-serving-area SA-List1
tai-list TAI-LIST2
!
```

Execute the following command to display **upf-info-list** configuration:

```
admin@# show running-config upf service-construct upf-info-list
upf service-construct upf-info-list THUD
upf-info BAR
...
!
!
```

**Note:** Multiple upf-info objects can be tied under a upf-info-list.  
Multiple upf-info-lists can be configured.

## Upf-service Configuration

Execute the following command to display upf-service configuration:

```
admin@# show running-config upf service | include upf-info
upf service primary-upf-info FOO
upf service upf-info-list THUD
```

**Note:** Parameters configured under FOO that is tied to **primary-upf-info** is published in upfInfo IE for UPF registration with NRF (including iwk-eps-ind, ue-ip-addr-ind, smf-serving-area, snssai-upfinfo-list).  
Only the priority and tai-list of upf-info objects configured under THUD upf-info-list are published in upfInfoList IE for UPF registration with NRF.

# Preparing sriovdpConfigMap in No OPM VM Deployment

## Prerequisite

- VFs from the same PFs need to appear on k8s under the same resource label. For example, in a 2 x 25G Intel 710 NIC, VFs from the first 25G port should be labeled **sr1** while VFs from the second 25G port should be labeled **sr2**.
- Make sure that all k8s nodes supporting the data plane have identical SRIOV configuration.
  - VFs are attached to the VM in the same order. For example, if the selected order for VFs is sr1, sr2, sr1, sr2 where the 2xsr1 VFs are from the first PF 25G port and the 2xsr2 VFs are from the second PF 25G port, then all VMs of this deployment need to have the sr1, sr2, sr1, sr2 port mapping.
  - A different deployment could select sr1, sr1, sr2, sr2 in which case all VMs of this deployment need to have the sr1, sr1, sr2, sr2 port mapping.

## Verification

- On the physical node, use **virsh list** and, on each of the VM, **virsh dumpxml | grep nova:name** to identify the data plane-supporting VMs by name. For example:  

```
<nova:name>cni-k8-node-56-io69sc-9gu39tp7t1m52551</nova:name>
```
- On the data plane-supporting VMs, run the following command. See in black the VM's PCI addresses for the SRIOV VFs and in green, their MAC addresses:

Figure A-1 VM PCI Address

```
[root@ccqe-r740v2-comp-2 ~]# virsh dumpxml 8 | grep -i vfio -A 8 -B 1 | grep -v -e source -e vlan -e tag -e alias
<mac address='fa:16:3e:4c:b4:13' />
<driver name='vfiop' />
<address type='pci' domain='0x0000' bus='0x13' slot='0x04' function='0x7' />
<address type='pci' domain='0x0000' bus='0x08' slot='0x06' function='0x8' />
--
<mac address='fa:16:3e:b1:b7:34' />
<driver name='vfiop' />
<address type='pci' domain='0x0000' bus='0x13' slot='0x08' function='0x5' />
<address type='pci' domain='0x0000' bus='0x08' slot='0x07' function='0x8' />
--
<mac address='fa:16:3e:94:ed:23' />
<driver name='vfiop' />
<address type='pci' domain='0x0000' bus='0x13' slot='0x04' function='0x5' />
<address type='pci' domain='0x0000' bus='0x08' slot='0x08' function='0x8' />
--
<mac address='fa:16:3e:d0:df:7b' />
<driver name='vfiop' />
<address type='pci' domain='0x0000' bus='0x13' slot='0x08' function='0x4' />
<address type='pci' domain='0x0000' bus='0x08' slot='0x09' function='0x8' />
```

- Use **ip link show** to identify the PFs to which the above MAC addresses belong. For example:

**Figure A-2 Identify the PFs**

```

6: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000
 link/ether e4:43:4b:79:95:90 brd ff:ff:ff:ff:ff:ff
 vf 0 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 1 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 2 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 3 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 4 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 5 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 6 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 7 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 8 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 9 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 10 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 11 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 12 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 13 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 14 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 15 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 16 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 17 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 18 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 19 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 20 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 21 MAC fa:16:3e:94:ed:23, spoof checking on, link-state auto, trust off
 vf 22 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 23 MAC fa:16:3e:4c:b4:13, spoof checking on, link-state auto, trust off

7: eno2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000
 link/ether e4:43:4b:79:95:91 brd ff:ff:ff:ff:ff:ff
 vf 0 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 1 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 2 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 3 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 4 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 5 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 6 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 7 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 8 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 9 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 10 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 11 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 12 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 13 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 14 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 15 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 16 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 17 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 18 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 19 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 20 MAC fa:16:3e:d0:df:7b, spoof checking on, link-state auto, trust off
 vf 21 MAC fa:16:3e:b1:bf:34, spoof checking on, link-state auto, trust off
 vf 22 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off
 vf 23 MAC 00:00:00:00:00:00, spoof checking on, link-state auto, trust off

```

The above use case is mapped as sr1, sr2, sr1, sr2:

- fa:16:3e:4c:b4:13 > 00:06.0 > eno1 first 25G port
- fa:16:3e:b1:bf:34 > 00:07.0 > eno2 second 25G port
- fa:16:3e:94:ed:23 > 00:08.0 > eno1 first 25G port
- fa:16:3e:d0:df:7b > 00:09.0 > eno2 second 25G port

## Config Maps

The configmap associated with the sr1, sr2, sr1, sr2 use case above is:

```

sriovdpConfigMap:
 resourceList:
 - resourceName: "sr1"
 selectors:
 drivers:
 - "vfio-pci"
 pciAddresses:
 - "0000:00:06.0"
 - "0000:00:08.0"

```

```

- resourceName: "sr2"
 selectors:
 drivers:
 - "vfio-pci"
 pciAddresses:
 - "0000:00:07.0"
 - "0000:00:09.0"

```

While a configmap associated with the sr1, sr1, sr2, sr2 use case is:

```

sriovdpConfigMap:
 resourceList:
 - resourceName: "sr1"
 selectors:
 drivers:
 - "vfio-pci"
 pciAddresses:
 - "0000:00:06.0"
 - "0000:00:07.0"
 - resourceName: "sr2"
 selectors:
 drivers:
 - "vfio-pci"
 pciAddresses:
 - "0000:00:08.0"
 - "0000:00:09.0"

```

PCI addresses above are only an example and need to be adapted for each deployment.





## Glossary

---

### 1

---

#### **10GBASE-BX4**

Common name of the standard that defines 10 gigabyte Ethernet data link over eight pairs of copper, four pairs for transmit and four pairs for receive.

---

#### **1Q**

See DOT1Q.

---

#### **1xRTT**

CDMA2000 Radio Transmission Technology

---

### 2

---

#### **2G**

Second generation wireless mobile telecommunications technology

---

### 3

---

#### **3DES**

Triple DES Technique where DES is performed multiple times to increase security level.

---

#### **3G**

Third generation wireless mobile telecommunications technology

---

#### **3GPP**

Third Generation Partnership Project – produce globally applicable technical specifications for 3G mobile systems based on evolved GSM.

---

### 4

---

#### **4G**

Fourth Generation wireless mobile telecommunications technology

---

### 5

---

#### **5G**

Fifth Generation wireless mobile telecommunications technology

---

---

**5G-NSA**

Fifth Generation Non-Standalone network

---

**5G-NSA NR**

Fifth Generation Non-Standalone network New Radio solution

---

**5G NR**

Fifth Generation New Radio network

---

**5G SA**

Fifth Generation Standalone

---

---

**A**

---

**A/AAAA**

A – Refers to the IPv4 address space. AAAA refers to the quad-A record – Refers to the IPv6 address space.

---

**AAA**

Authentication, Authorization, and Accounting – Key functions to intelligently control access, enforce policies, audit usage, and provide the information necessary to perform billing for services.

---

**AAFA**

Active Flow Analytics

---

**AAR**

AA-Request

---

**ABNF**

augmented Backus–Naur form

---

**ABQP**

Aggregate BSS QoS Profile

---

**ABR**

Area Border Router

---

**ACL**

Access Control List

---

**ACR**

Anonymous Customer Reference – an encrypted MSISDN.

---

**ACS**

Access Controller Server

---

**ADC**

Analog-to-Digital Conversion

---

**ADMF**

Administration Function/Administration Function for Legal Intercept

---

---

**AdvancedTCA**

Advanced Telecommunications Computing Architecture

---

**AE**

Application Entity

---

**AES**

Advanced Encryption Standard A symmetric key encryption technique.

---

**AF**

Access Function

---

**AFC**

Active Failover Cluster – Consists of nodes that are all active and ready to provide services channeled through a load-balancer/router.

---

**AGW**

Access Gateway

---

**AIS**

Application Interface Specification

---

**AJP**

Apache JServ Protocol – Packet-based protocol that allows HTTP Web Server to communicate with Servlet Engine instances.

---

**AKA**

Authentication and Key Agreement

---

**ALG**

Application Layer Gateway

---

**AMBR**

Aggregated Maximum Bit Rate

---

**AMC**

Advanced Mezzanine Card

---

**AN-SCEF**

Affirmed Networks Service Capability Exposure Function VNF that implements the 3GPP SCEF+MTC-IWF functions specified in 3GPP 23.682[1].

---

**ANSI**

American National Standards Institute

---

**AOC**

Advice of Charge

---

**APCO**

Additional Protocol Configuration Option

---

**API**

Application Programming Interface

---

---

**APN**

Access Point Name – gateway name between a GPRS, 3G, or 4G mobile network and the public Internet. A mobile device making a data connection must be configured with an APN to present to the carrier. APN identifies the packet data network (PDN) with which a mobile data user wants to communicate.

---

**APN-AMBR**

Access Point Name - Aggregated Maximum Bit Rate

---

**APN-NI**

APN Network Identifier

---

**APN-OI**

APN Operator Identifier

---

**APS**

Automatic Protection Switching

---

**AR**

Re-Auth-Request

---

**ARP**

Address Resolution Protocol/Allocation and Retention Policy/Allocation and Retention Priority

---

**AS**

Application Server

---

**ASAP**

Affirmed Service Automation Platform

---

**ASM**

Advanced Services Module – A VM instance that runs the tasks needed for providing advanced services like HTTP Proxy, Cache, WAP GW, etc.

---

**ASN.1**

Abstract Syntax Notation One

---

**ASP**

Application Server Process, as defined by RFC 4666

---

**ASR**

Abort-Session-Request

---

**Asset**

Refers to vBuilder vendor objects, vendor components, service components, and service models.

---

**Association Service Component**

Used by a service model to define the relationship between a “From” service component and a “To” service component in an association. It can be a regular service component that represents associated objects for specific devices based on their schema, or it can be an ‘empty’ service component that only represents the relationship between the service components but not the associated objects on the vendor object instance/device instance.

---

**ATCA**

See AdvancedTCA.

---

**ATM**

Asynchronous Transfer Mode

---

**AUSF**

Authenticated Server Function

---

**AVOD**

Advertisement Video on Demand

---

**AVP**

Attribute Value Pair

---

**AWG**

American Wire Gauge

---

**AWT**

Average Working Time

---

---

**B**

---

**B-Frame**

Bidirectionally-predicted Frame – Opens up the ability for the encoder to send frames in an arbitrary order, so the decoder may need to decode multiple frame before displaying them. Frames can be predicted both backward time and forward time (forward referenced frames need to be sent out of order)

---

**BBERF**

Bearer Binding and Event Reporting Function

---

**BBS**

Basic Blade Service

---

**BCD**

binary-coded decimal – a family of representations of numerals, uppercase Latin letters, and some special and control characters as six-bit character codes.

---

**BER**

Basic Encoding Rules

---

**BFD**

Bidirectional Flow Detection/Bidirectional Forwarding Detection

---

**BGP**

Border Gateway Protocol

---

**BH**

Busy Hour

---

**BIX**

Byte Information Exchange

---

---

**BS**

Base Station/Billing System

---

**BSC**

Base Station Controller – The BSC is in charge of the allocation and release of radio channels and handover management.

---

**BSN**

Broadband Service Node

---

**BSS**

Business Support System/Base Station System or Subsystem

---

**BSSAP**

Base Station System Application Part

---

**BSSGP**

Base Station System GPRS Protocol

---

**BT**

Block Transfer

---

**BVC**

BSSGP Virtual Connection – An end-to-end virtual communications path between remote network service user entities.

---

**BVCI**

BSSGP Virtual Connection Identifier

---

---

**C**

---

**C-IoT**

Cellular Internet of Things

---

**C-plane**

Control plane – Administration functions outside the transfer of user data packets associated with functions such as setting up and tearing down of connections, establishing identities, and authorizing users.

---

**C-SGN**

Cellular Serving Gateway Node/Cellular Internet of things (CIoT) Serving Gateway Node

---

**CALEA**

Communications Assistance for Law Enforcement Act

---

**CallP**

Call Processing, a type of Virtual Machine on the MCC MME

---

**CAMEL**

Customized Application for Mobile Network Enhanced Logic

---

**CAP**

CAMEL Application Part/Common Administration Part

---

<b>Capex</b>	Capital Expenditure
<b>CAR</b>	Common Alarm Record
<b>CATT</b>	Common Automated Test Tool
<b>CBC</b>	Cell Broadcasting Center
<b>CBR</b>	Constant Bit Rate – useful for streaming multimedia content on limited capacity channels, as it takes advantage of all available capacity.
<b>CC</b>	Content of Communications
<b>CCA</b>	Credit-Control-Answer message
<b>CCSID</b>	CC Link Identifier
<b>CCM</b>	Centralized Control Module – A VM instance that runs the centralized tasks needed for call-control functionality and routing protocols.
<b>CCR</b>	Credit-Control-Request message
<b>CDF</b>	Charging Data Function
<b>CDMA</b>	Code Division Multiple Access
<b>CDN</b>	Content Delivery Network
<b>CDP</b>	Common Delivery Part
<b>CDR</b>	Charging Data Record – is a formatted collection of information about a chargeable event (for example, time of call set-up, duration of the call, amount of data transferred, and so on) for use in billing and accounting. For each party to be charged for parts of or all charges of a chargeable event, a separate CDR is generated. Multiple CDRs may be generated for a single chargeable event, for example, because of its long duration, or because more than one charged party is to be charged. Also referred to as Call Data Record or Call Detail Record
<b>CFC</b>	Cold Failover Cluster Consists of one primary node that serves requests and one or more failover or standby nodes. At any given time, only one node in the cluster is alive and



---

manages all system resources.

---

**CFDS**

Customer Feature Delivery System

---

**CG**

Charging Gateway – The device on the GSM GPRS or UMTS network that collects and maintains Call Detail Records (CDRs) for subscriber PDP contexts. Also referred to as a Charging Gateway Function (CGF).

---

**CGF**

Charging Gateway Function – See CG.

---

**CGI**

Cell Global Identity

---

**CHAP**

Challenge Handshake Authentication Protocol

---

**CIA**

Configuration Information Answer

---

**CID**

Communication Identifier

---

**CII**

Communication Identifying Information

---

**CIN**

Communication Identity Number

---

**CIoT**

Cellular Internet of Things

---

**CIQ**

Customer Information Questionnaire

---

**CIR**

Configuration Information Request

---

**CK/IK**

Cipher Key/Integrity Key

---

**CLI**

Command Line Interface

---

**CM**

Configuration Management

---

**CMA**

Connection Management Answer

---

**CMAS**

Commercial Mobile Alert System messages

---

**CMR**

Connection Management Request

---

---

<b>CN</b>	Compute Node/Core Network
<b>CNA</b>	Cloud Native Architecture
<b>CoA</b>	Change of Authorization
<b>Coalescing</b>	A grouping of parameters that share the same base type object class and are configured with the same values. This feature simplifies configuration and reduces data entry errors. The value supplied to a coalesced parameter is propagated to all child parameters.
<b>codec</b>	Device or computer program capable of encoding or decoding a digital data stream or signal.
<b>Compression Gain</b>	Ratio of bytes-in to bytes-out.
<b>CoS</b>	Classes of Service
<b>COTS</b>	Commercial Off The Shelf
<b>CP</b>	Control Processor – Control processors that manage and control applications supporting both system and network functions.
<b>CPA</b>	Certification Path Advertisement
<b>CPE</b>	Control Plane Engine
<b>CPF</b>	Control Plane Function
<b>CPM</b>	Control Plane Module
<b>CPR</b>	Common Performance Record
<b>CPS</b>	Certification Path Solicitation
<b>CPU</b>	Central Processing Unit
<b>CRC</b>	Cyclic Redundancy Check – The monitoring of a digital bit stream to detect deviations from the expected bit patterns.

---

**CRR**

Call Release Reason

---

**CS**

Circuit Switched/Corporate Standards

---

**CSCF**

Call State Control Function

---

**CSFB**

Circuit Switched Fallback

---

**CSG**

Closed Subscriber Group

---

**CSGID**

Closed Subscriber Group Identity

---

**CSI**

CAMEL Subscription Information

---

**CSL**

Call Summary Log

---

**CSM**

Control/Content Services Module – a VM instance that runs all services provided by CPM and ASM.

---

**CSR**

create session request

---

**CSV**

Comma Separated Value

---

**CUPS**

Control and User Plane Separation

---

**D**

---

**DASH**

Dynamic Adaptive Streaming over HTTP

---

**Data Hiding**

Filtering out unused (optional) parameters and parameters whose values have already been set. Data hiding is accomplished by applying the Hard Code or Ignore decoration to a parameter.

---

**DB**

Database Database files (such as control files, redo-log files, and data files) and a set of processes that manage them along with various system mechanisms (such as shared memory and semaphores).

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---

<b>DC</b>	Direct Current/Diameter Client
<b>DCCA</b>	Diameter Credit-Control Application
<b>DCM</b>	Distributed Control Module
<b>DCNR</b>	Dual Connectivity with New Radio
<b>DDR</b>	Double Data Rate
<b>DEA</b>	Diameter-EAP-Answer – command sent by the Diameter server to the client.
<b>DER</b>	Diameter-EAP-Request – command sent by a Diameter client to a Diameter server that conveys an EAP-Response from the EAP client.
<b>DES</b>	Data Encryption Standard A symmetric key encryption technique.
<b>Device Instance</b>	An instance of an actual device, e.g., a device, application, tool, etc. that is registered in and known by vTransactor.
<b>Device Type</b>	A generic term used to define one or more vendor devices, network elements (NE), or applications, such as, Affirmed MCC, MCC EMS, and Cisco PCRF.
<b>DF</b>	Delivery Function/Delivery Function for Legal Intercept
<b>DFA</b>	Data Flow Analysis
<b>DG</b>	Device Gateway
<b>DHCP</b>	Dynamic Host Configuration Protocol
<b>DIMM</b>	Dual In-line Memory Module
<b>DMZ</b>	Demilitarized Zone – Refers to one or more computers that are outside the firewall in a networking context. These computers/servers broker requests from outside (Internet) as well as the rest of the local area network.
<b>DNS</b>	Domain Name System – A system that stores information associated with domain names in a distributed database on networks, such as the Internet. Part of the IP protocol stack, this

---

	allows host name to IP address mapping within an Internet environment. Also, Domain Name Service
<b>DOIC</b>	Diameter Overload Indication Conveyance
<b>DoNAS</b>	Data over Non Access Stratum
<b>DOT1Q</b>	IEEE 802.1q – the networking standard that supports virtual LANs (VLANs) on an Ethernet network.
<b>Downlink Traffic</b>	HSDPA traffic traveling from the RNC to the UE.
<b>DP</b>	Discard Priority
<b>DPDK</b>	Data Plane Development Kit – a set of data plane libraries and NIC drivers for fast packet processing.
<b>DPE</b>	Data Plane Engine
<b>DPFW</b>	Data Plane Framework
<b>DPI</b>	Deep Packet Inspection
<b>DPR</b>	Disconnect Peer Request
<b>DRA</b>	Delivery Report Answer/DHCP Relay Agent/Diameter Routing Agent
<b>DRA/SLF</b>	Diameter Routing Agent/Subscriber Location Function
<b>DRM</b>	Digital Rights Management
<b>DRMP</b>	Diameter Message Priority
<b>DRR</b>	Delivery Report Request
<b>DRT</b>	Data Record Transfer
<b>DSCP</b>	Differentiated Services Code Point – A field in the IPv4/6 headers that describes the desired QoS of a packet.

---

**DST**  
Daylight Savings Time

---

**DTA**  
Device Trigger Answer

---

**DTR**  
Device Trigger Request

---

**DUPMS**  
Distributed UDP Path Management System

---

**DUT**  
Device Under Test

---

**DVL**  
Data Volume Limit

---

**DVLT**  
Data Volume Limit Trigger

---

**DWA**  
Device-Watchdog-Answer

---

**DWR**  
Device-Watchdog-Request

---

## **E**

---

**E-RAB**  
E-UTRAN Radio Access Bearer

---

**E-SMLC**  
Evolved Serving Mobile Location Center

---

**E-UTRAN**  
Evolved UMTS Terrestrial Radio Access Network

---

**E-UTRAN Node B**  
Evolved Node B

---

**E/W**  
East/West traffic

---

**eAN**  
Evolved Access Node for CDMA HRPD network

---

**EAP**  
Extensible Authentication Protocol

---

**EAP-AKA**  
Extensible Authentication Protocol - Authentication and Key Agreement

---

**EAP-GTC**  
Extensible Authentication Protocol - Generic Token Card

---

---

<b>ECC</b>	Enterprise Cloud Communications/Embedded Communications Computing
<b>ECDSA</b>	Elliptic Curve Digital Signature Algorithm – offers a variant of the Digital Signature Algorithm (DSA) which uses elliptic curve cryptography.
<b>ECGI</b>	E-UTRAN Cell Global Identity
<b>ECM</b>	EPC Connection Management
<b>EDP</b>	Event Detection Point
<b>EDR</b>	Event Data Record
<b>EEPROM</b>	Electrically Erasable Programmable Read-only Memory
<b>EGP</b>	Exterior Gateway Protocol
<b>EGS</b>	Entitlement Generation Server
<b>EIR</b>	Equipment Identity Register
<b>EJB</b>	Enterprise JavaBeans
<b>ELP</b>	EPC LCS Protocol
<b>EMC</b>	Electromagnetic Compatibility
<b>EMI</b>	Electromagnetic Interference
<b>EMM</b>	EPS Mobility Management
<b>EMS</b>	Element Management System/Enhanced Messaging Service
<b>EN</b>	European Norm
<b>eNB</b>	Evolved Node B
<b>eNodeB</b>	Evolved Node B

---

**EP**

Emission Priority

---

**EPC**

Evolved Packet Core – Provides a converged voice and data-networking framework to connect users on a Long-Term Evolution (LTE) network. Also, Enhanced Packet Core

---

**ePDG**

Evolved Packet Data Gateway

---

**EPG**

Electronic Program Guide

---

**EPLMN**

Equivalent PLMN

---

**EPS**

Evolved Packet System

---

**eRANAP**

enhanced RANAP (Radio Access Network Application Protocol)

---

**eSCEF**

Affirmed Networks enhanced Service Capability Exposure Function product that is a composite VNF (cVNF) that instantiates the DG and the AN-SCEF VNFs to provide a complete API based IOT solution (for transport and other services such as monitoring) for service providers.

---

**ESD**

Electrostatic Discharge

---

**ESME**

External Short Messaging Entities

---

**ESP**

Encapsulated Security Payload

---

**ETRAN**

Enhanced Terrestrial Radio Access Network

---

**ETSI**

European Telecommunication Standards Institute

---

**EV-DO**

EV-Data Optimized

---

**EVOD**

Exclusive Video on Demand

---

---

**F**

---

**FACH**

Forward Access Channel

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<b>FCC</b>	Federal Communications Commission
<b>FD</b>	Functional Description
<b>FFS</b>	For Future Study
<b>FGW</b>	Femtocell Gateway, also called small cell gateway
<b>FIFO</b>	First In, First Out
<b>FM</b>	Fault Management
<b>Footprint</b>	Represents the collection of devices or applications (device instances) that are affected when you roll out a snapshot. See also Snapshot (page 30).
<b>FP</b>	Fastpath/Framing Protocol/Functional Processor
<b>FPGA</b>	Field Programmable Gate Array
<b>FQDN</b>	Fully Qualified Domain Name
<b>FQPC</b>	Fully Qualified Partial CDR
<b>FRS</b>	Feature Requirements Specification/Functional Requirements Specification
<b>FRU</b>	Field Replaceable Unit
<b>FSM</b>	Finite State Machine
<b>FTP</b>	File Transfer Protocol
<b>FTS</b>	Feature Technical Specification
<b>FUBM</b>	Fair Use Bandwidth Management
<b>FUI</b>	Final-Unit-Indication
<b>FVOD</b>	Free Video on Demand

---

## **G**

---

### **Ga**

Interface serving the CDRs that are written in the GSN and sent to the charging gateway (CG). The Ga interface uses a GTP-based protocol, with modifications to support CDRs (Called GTP'). The interface is between the SGSN and the CGF.

---

### **Gb**

Interface between the base station subsystem BSS and the SGSN. The transmission protocol can be Frame Relay or IP.

---

### **GB**

Gigabyte

---

### **Gbps**

Gigabits per second

---

### **GBR**

Guaranteed Bit Rate

---

### **GCSNA**

Circuit Services Notification Application

---

### **Gd**

Interface between the SGSN and the SMS Gateway. The Gd interface can use MAP1, MAP2 or MAP3. The Gd interface is between the SGSN and the GSM SMS-GMSC/SMS-IW MSC.

---

### **Ge**

Interface between the GSN and the service control point (SCP). Uses the CAP protocol. The Ge interface is between the SGSN and the EIR.

---

### **GEC**

Global Emergency Call

---

### **GERAN**

GPRS EDGE Radio Access Network

---

### **Gf**

Interface between the SGSN and the Equipment Identity Register (EIR). It is used for checking the International Mobile Equipment Identity (IMEI) against a list of reported stolen mobile phones.

---

### **GGSN**

Gateway GPRS Support Node – Interface between the GPRS wireless data network and other networks such as the Internet or private networks. It supports the edge routing function of the GPRS network. The GGSN is the node within the GPRS core network that is responsible for inter-working with external PDNs, such as the Internet.

---

### **Gi**

1. IP based interface between the GGSN and a public data network (PDN), either directly to the Internet or through a WAP gateway. 2. An external interface between the GGSN and

---

	another packet data network.
<b>GI</b>	Geographical Identity
<b>Gi-LAN</b>	A 3GPP term for a network upstream from the PGW. In Gi-LAN, value-added services are executed including TDF, firewall, Carrier-grade NAT, TCP optimization, traffic optimization, and URL filtering.
<b>GiGW</b>	Gi Gateway. See Gi
<b>Gmb</b>	Interface between the GGSN and the Broadcast-Multicast Service Center (BM-SC) that is used for controlling MBMS bearers.
<b>GMLC</b>	Gateway Mobile Location Center
<b>GMM</b>	GPRS Mobility Management
<b>GMM context</b>	GPRS Mobility Management – The GMM context is the information sets held at the MS and the SGSN. It contains mobility-related information such as the Mobility Management state as well as location information, IMSI, and negotiated timer values.
<b>Gn</b>	IP-based interface between the SGSN and other SGSNs and (internal) GGSNs. The interface uses the GTP Protocol. DNS also shares this interface.
<b>GNP</b>	Generic Network Product – a class of network products that all implement a common set of 3GPP-defined functionalities for that particular network product.
<b>Gp</b>	IP-based interface between the internal SGSN and external GGSNs. The interface uses the GTP Protocol.
<b>GPP</b>	3rd Generation Partner Project
<b>GPRS</b>	General Packet Radio Service – A packet radio access technique based on GSM radio to transfer data in an efficient manner, optimizing the use of network resources. The GPRS re-uses existing GSM radio technology.
<b>GPS</b>	Global Product Support
<b>Gr</b>	Interface between the SGSN and the HLR. Messages going through this interface use the MAP3 protocol.

---

---

**GRE**

Generic Routing Encapsulation

---

**Gs**

Interface between the SGSN and the MSC (VLR). The interface uses the BSSAP+ protocol. The Gs interface allows paging and station availability when it performs data transfer. The Gs interface provides mobile stations the ability to communicate with the MSC/VLR for circuit switch services while they are attached to the packet network.

---

**GSM**

Global System for Mobile communications – A technical standard for European mobile networks.

---

**GSN**

GPRS Support Node – Constitutes the interface between the radio system and the fixed networks for packet switched services. The GSN performs all necessary functions in order to handle the packet transmission to and from the mobile stations. The GPRS Support Node can be either SGSN or GGSN.

---

**GSU**

Granted Service Units (Container)

---

**GT**

Global Title

---

**GTP**

GPRS Tunneling Protocol – The transport protocol used between the SGSN and GGSN for both signaling and user data transfer. (C-Plane or U-Plane)

---

**GTP-C**

GPRS Tunneling Protocol - C-Plane – The Protocol for the control plane handling of signaling between the GSNs within the core network. GPRS Tunnel Protocol-Control is the protocol between the subscriber control and the GGSN used to communicate on behalf of the GMM and SM functions.

---

**GTP-U**

GPRS Tunneling Protocol - U-Plane – The Protocol used for carrying user data within the GPRS core network and between the Radio Access Network and the core network.

---

**GTPP**

GTP Prime Protocol

---

**GTT**

Global Title Translation

---

**GUMMEI**

Globally Unique Mobility Management Entity Identifier

---

**GUTI**

Globally Unique Temporary Identity

---

**GW**

Gateway

---

---

**Gx**

Off-line policy interface between the GGSN and the charging rules function (CRF). It is used for provisioning the service data flow based on charging rules. The interface uses the diameter protocol.

---

**Gy**

On-line charging interface between the GGSN and the online charging system (OCS). The interface uses the diameter protocol (DCCA application).

---

**Gz**

Off-line (CDR-based) charging interface between the GSN and the CG. The interface uses GTP'.

---

---

**H**

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**HA**

High Availability

---

**Hello Packet**

In the Open Shortest Path First (OSPF) communications protocol that enables network routers to share information with each other, a Hello packet is a special packet (message) that a router sends out periodically to establish and confirm network adjacency relationships.

---

**Hello Protocol**

OSPF creates adjacencies between neighboring routers to exchange routing information. Not every two neighboring routers will become adjacent. The Hello Protocol is responsible for establishing and maintaining neighbor relationships. It also ensures bidirectional communication between neighbors. HELLO packets are sent periodically out all router interfaces. Bidirectional communication is indicated when the router sees itself listed in the neighbor's HELLO packet.

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**HFN**

Hyper Frame Number

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**HI**

Handover Interface

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**HI1**

Handover Interface Port 1 (for Administrative Information)

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**HI2**

Handover Interface Port 2 (for Intercept-related Information)

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**HI3**

Handover Interface Port 3 (for Content of Communication)

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**HLR**

Home Location Register – Used to store the subscription record, temporary data, the serving-MSC number and VLR number of a mobile subscriber.

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**HLS**

HTTP Live Streaming

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**HPLMN**

Home PLMN – This represents the operator of an incoming roamer.

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**HRPD**

High Rate Packet Data, also referred to as 1xEV-DO for high speed data over CDMA.

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**HSS**

Home Subscriber Server – The concatenation of the HLR (Home Location Register) and the AuC (Authentication Center). The HLR part of the HSS stores and updates the database containing all subscription information, when necessary.

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**HTDR**

HTTP Transaction Data Record

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**HTTP**

HyperText Transport Protocol

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**I**

---

**I/O**

Input/Output

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**IAP**

Intercept Access Point

---

**ICD**

Interface Control Document

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**ICE**

Interception Control Elements

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**ID**

Identifier

---

**IDi**

Identification:Initiator – an identification payload that allow peers to assert an identity to one another. In the case of the ePDG, the IDi is always the UE.

---

**IDr**

Identification:Responder – an identification payload that allow peers to assert an identity to one another. In the case of the ePDG, the IDr is always the ePDG.

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**IDR**

Insert Subscriber Data Request

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**IE**

Information Element

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**IEC**

International Electrotechnical Commission

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**IEEE**

Institute of Electrical and Electronics Engineers

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<b>IKE</b>	Internet Key Exchange – Control plane protocol used by IP Security Protocol (IPsec).
<b>ILF</b>	IMSI Locator Function
<b>IM</b>	Installation Method
<b>IMC</b>	IPMC Master Controller
<b>IMEI</b>	International Mobile Equipment Identity
<b>IMEISV</b>	International Mobile Equipment Identity of the equipment served/International Mobile Equipment Identity Software Version
<b>IMS</b>	IP Multimedia Subsystem/IP Multimedia Services/International Mobile Subscriber
<b>IMSI</b>	International Mobile Subscriber Identity
<b>IN</b>	Intelligent Network
<b>IOM</b>	I/O Module – a VM instance that is responsible for receiving packets into the MCC and sending the packets out.
<b>IOS</b>	Inter-Operability Specification
<b>IoT</b>	Internet of Things
<b>IP</b>	Internet Protocol – One of the fundamental protocols of the TCP/IP Internet Protocol suite. It defines the IP datagram as the unit of information passed across an Internet and provides the basis for connection-less, best-effort packet delivery service.
<b>IPC</b>	Inter-Process Communication
<b>IPMB</b>	Intelligent Platform Management Bus
<b>IPMC</b>	Intelligent Platform Management Controller/Intelligent Peripheral Management Controller
<b>IPMI</b>	Intelligent Platform Management Interface

---

<b>IPsec</b>	IP Security Protocol
<b>IPSP</b>	IP Server Process as defined by RFC 4666. IPSP is essentially the same as ASP. For the SGSN, IPSP is used to denote the function interfaces with the RNC, whereas ASP is used to denote the function interfaces with the core SS7 network.
<b>IRAT</b>	Inter Radio Access Type
<b>IRAU</b>	Inter-SGSN Routing Area Update – Procedure that enables a subscriber to move between cells covered by different SGSNs while maintaining an active PDP context.
<b>IRI</b>	Intercept Related Information
<b>ISC</b>	IPMC Slave Controller
<b>ISD</b>	Insert Subscriber Data
<b>ISM</b>	Integrated Services Module
<b>ISO</b>	International Organization for Standardization
<b>ITU</b>	International Telecommunication Union
<b>Iu</b>	General term for referring to 3GPP interface that links the RNC with the GGSN.
<b>IuCS</b>	3GPP circuit-switched interface that links the RNC with the MSC.
<b>IuPS</b>	3GPP packet-switched interface that links the RNC with the GGSN.
<b>IVOD</b>	Interactive Video on Demand – VOD with interactive features such as play, resume, stop, pause, fast forward, and with the ability to avoid advertisements, purchase goods, and so on.
<b>IWM</b>	Integrated Workflow Module
<b>IWMSC</b>	Interworking Mobile Switching Center
<b>IWS</b>	Interworking System



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## J

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### **Java Messaging Service (JMS)**

One of the J2EE components involved in relaying messages (point-to-point or publish-subscribe types) between Java objects via TCP/IP in the Java Messaging Service.

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### **JavaScript Object Notation (JSON)**

Data exchange format used in the request and response messages through the vTransactor northbound API calls.

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### **JMS**

Java Messaging Service – One of the J2EE components involved in relaying messages (point-to-point or publish-subscribe types) between Java objects via TCP/IP.

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## K

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### **KASME**

Key Access Security Management Entries

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### **KCI**

Key Control Indicator

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### **KNI**

Kernel NIC Interface

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### **KPI**

Key Performance Indicators

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### **KS**

Knowledge Services

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### **KSI**

Key Set Identifier

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### **KVM**

Kernel-based Virtual Machine

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## L

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### **L2oGRE**

Layer 2 over Generic Routing Encapsulation

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### **L2TP**

Layer 2 Tunneling Protocol

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### **LAI**

Location Area Identity

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<b>LAN</b>	Local Area Network
<b>LB</b>	Load Balancer
<b>LB VM</b>	Load Balancer Virtual Machine
<b>LBB</b>	Location Based Billing
<b>LBO</b>	Local Break-Out
<b>LCF</b>	Location Client Function
<b>LCS</b>	Location Services
<b>LCS-AP</b>	LCS Application Protocol
<b>LDAP</b>	Lightweight Directory Access Protocol
<b>LEA</b>	Law Enforcement Agency
<b>LED</b>	Light Emitting Diode
<b>LEMF</b>	Law Enforcement Monitoring Facility
<b>LFU</b>	Least Frequently Used
<b>Lg</b>	Interface between the SGSN and the Gateway Mobile Location Center (GMLC) that is used for location-based services.
<b>LI</b>	Lawful/Legal Intercept
<b>LI ID</b>	Lawful Intercept Identity
<b>LIAP</b>	Lawful Intercept Access Part/Legal Interception Access Protocol
<b>LICP</b>	Legal Interception Common Protocol
<b>LIG</b>	Lawful/Legal Intercept Gateway

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<b>LII</b>	Lawful Intercept Identifier
<b>LIID</b>	Lawful Interception Identifier
<b>LIPA</b>	Local IP Access
<b>LISP</b>	Lawful Interception Service Processor
<b>LLC</b>	Logical Link Control/Logical Link Control Layer Protocol
<b>LLC layer</b>	Logical Link Control – The upper sublayer of the data link layer of the seven-layer OSI model that provides media-independent functions and the logical connection between the stations within the local area network.
<b>LMS</b>	License Management System
<b>LP</b>	Logical Processor
<b>LPM</b>	Longest Prefix Match
<b>LPP</b>	LTE Positioning Protocol
<b>LPWAN</b>	Low Power Wide Area Network
<b>LRU</b>	Least Recently Used
<b>LSA</b>	Link State Advertisement – Unit of data describing the local state of a router or network. For a router, it includes the state of the router's interfaces and adjacencies. Each link state advertisement is flooded throughout the routing domain. The collected link state advertisements of all routers and networks forms the protocol's link state database.
<b>LSAF</b>	Location Subscriber Authorization Function
<b>LSB</b>	Least Significant Bit
<b>LSBF</b>	Location System Billing Function
<b>LSOF</b>	Location System Operations Function

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**LSPF**  
Location Subscriber Privacy Function

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**LTE**  
Long Term Evolution

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**LTMI**  
Long Term Mobile ID

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**LTS**  
Long Term Support

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**LX**  
Long Range Optical Transceiver

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## **M**

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**M-CDR**  
Mobility CDR

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**M-TMSI**  
MME-Temporary Mobile Subscriber Identity

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**M2M**  
Machine-to-Machine

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**M3UA**  
MTP Level 3 (MTP3) User Adaptation Layer

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**MAC**  
Medium Access Control

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**MAP**  
Mobile Application Part – The MAP is the software protocol layer that allows information about a roaming mobile-subscriber (MS) to flow between functional entities in the Public Land Mobile Network (PLMN).

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**MB**  
Megabyte

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**MBMS**  
Multimedia Broadcast-Multicast Service

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**Mbps**  
Megabits per second

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**MBR**  
Maximum Bit Rate

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**MBSFN**  
MBMS over a Single Frequency Network

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**MCA**  
Management Configuration Adapter

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<b>MCC</b>	Mobile Content Cloud / Mobile Country Code
<b>MCD</b>	Messaging Channel Database
<b>MCE</b>	Multi-cell/Multicast Coordination Entity
<b>MCG</b>	Master Cell Group
<b>MCM</b>	Management Configuration Module
<b>MD</b>	Mediation Device
<b>ME</b>	Mobile Entity/Managed Element/Mobile Equipment
<b>MeNB</b>	Master eNB
<b>MF</b>	Mediation Function
<b>MH-BGP</b>	Multi-homing Border Gateway Protocol
<b>MiTM</b>	Man-in-the-middle-attack – when an attacker secretly relays and possibly alters the communication between two parties who believe they are directly communicating with each other.
<b>MLT</b>	Multi Link Trunk
<b>MM</b>	Mobility Management
<b>MMC</b>	Mezzanine Management Controller
<b>MME</b>	Mobility Management Entity
<b>MMS</b>	Multimedia Messaging Service
<b>MMSC</b>	Multimedia Messaging Service Center
<b>MNC</b>	Mobile Network Code

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<b>MNO</b>	Mobile Network Operator
<b>MO</b>	Managed Object/Mobile Originated
<b>MO-LR</b>	Mobile Originated Location Request
<b>MOBIKE</b>	IKEv2 Mobility and Multi-homing Protocol – for more information, see RFC 4555.
<b>MONTE</b>	Monitoring Enhancements
<b>MPC</b>	Modular Port Concentrators
<b>MPLS</b>	Multi-protocol Label Switching
<b>MPU</b>	Main Processing Unit
<b>MRA</b>	Multi-protocol Routing Agent
<b>MRM</b>	Management and Resource Module
<b>MS</b>	Mobile Station – Subscriber equipment that provides access into the Public Land Mobile Network (PLMN).
<b>MSC</b>	Mobile Switching Center – Provides switching and call control of GSM basic and supplementary services.
<b>MSC/VLR</b>	Mobile Switching Center/Visitor Location Register
<b>MSCC</b>	Multiple Services Credit Control
<b>MSF</b>	Mediation Service Function
<b>MSISDN</b>	Mobile Subscriber Integrated Services Digital Network-Number – A number string used to uniquely identify a mobile telephone subscriber in a GSM or a UMTS mobile network.
<b>MSM</b>	Mobile Service Module
<b>MSS</b>	Maximum Segment Size/Multi-service Switch

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<b>MT</b>	mobile terminated
<b>MT-LR</b>	Mobile Terminated Location Request
<b>MT TDA</b>	Mobile Terminated Data-Answer
<b>MT TDR</b>	Mobile Terminated Data Request
<b>MTC</b>	Machine-Type Communications
<b>MTC-IWF</b>	Machine Type Communications-InterWorking Function
<b>MTU</b>	Maximum Transmission Unit
<b>MVNE</b>	Mobile Virtual Network Enabler
<b>MVNO</b>	Mobile Virtual Network Operator
<b>MZ</b>	Militarized Zone – A local area network or a sub network behind a firewall that is not directly accessible by any computer on the Internet.
<b>N</b>	
<b>NA</b>	Not Applicable/Neighbor Advertisement
<b>NACC</b>	Network Assisted Cell Change
<b>NAI</b>	Network Access Identifier
<b>NAPT</b>	Network Address Port Translator
<b>NAPTR</b>	Naming Authority Pointer Resource Record is defined by the IETF for Dynamic Delegation Discovery System (DDDS). It is defined in RFC 3401, 3402, 3403 and 3404.
<b>NAS</b>	Non Access Stratum/Network Attached Storage
<b>NAT</b>	Network Address Translation

---

**NB**

narrowband

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**NB-IoT**

Narrow Band – Internet of Things (aka Narrowband-IoT) – an LPWAN radio technology standard that has been developed to enable a wide range of devices and services to be connected using cellular telecommunications bands.

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**NBA**

Node Backbone Area

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**NB-IoT**

Narrow Band – Internet of Things (aka Narrowband-IoT) – an LPWAN radio technology standard that has been developed to enable a wide range of devices and services to be connected using cellular telecommunications bands.

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**NBR**

NAT Binding Records

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**ND**

Neighbor Discovery

---

**NDP**

Neighbor Discovery Protocol

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**NE**

Network Element

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**NE-RAMS**

Network Element Reliability, Availability, Maintainability, and Survivability

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**NEAF**

Non-EPS Alert Flag

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**NEBS**

Network Equipment Building System

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**NFA**

Network Flow Analysis

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**NFV**

Network Functions Virtualization – Allows a network operator to deploy network functions as virtualized software instances instead of dedicated hardware appliances.

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**NFV MANO**

Network Functions Virtualization Management and Organization

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**NFVI**

NFV Infrastructure – Entire NFV environment across all locations consisting of both hardware and software components.

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**NI-LR**

Network Induced Location Request

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**NIA**

Non-IP Data Delivery (NIDD) Information Answer

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<b>NIC</b>	Network Interface Card
<b>NID</b>	Network Identifier
<b>NIDD</b>	Non-IP Data Delivery
<b>NIR</b>	Non-IP Data Delivery (NIDD) Information Request
<b>NMIS</b>	Network Management Interface Specification
<b>NOR</b>	Notify Request
<b>northbound</b>	Interfaces from function under discussion towards the enterprise/application servers.
<b>NPLI</b>	Network Provided Location Information
<b>NPU</b>	Network Processing Unit
<b>NRI</b>	Network Resource Identifier
<b>NS-VC</b>	Network Services Virtual Connection – An end-to-end logical connection for the transport of signaling and data packets between the SGSN and the BSS.
<b>NS layer</b>	Network Services layer – The transport protocol used between the SGSN and the BSS, specifically the PCUSN of the BSS, to ensure in-order delivery of packets sent between higher layer protocols.
<b>NSE</b>	Network Services Entity – A logical entity that exists on both the SGSN and the BSS for managing the transport of both signaling and data packets between them.
<b>NSSA</b>	Not-So-Stubby-Area
<b>NSWO</b>	Non-Seamless WLAN Offload
<b>NTP</b>	Network Time Protocol
<b>NTPd</b>	Network Time Protocol daemon

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**NUMA**

Non Uniform Memory Access

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**NVOD**

Near Video on Demand

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**O**

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**OA&M**

Operations, Administration and Maintenance – All the tasks necessary for providing, maintaining, or modifying the services provided by a switching system.

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**OAM**

Operations, Administration and Maintenance – All the tasks necessary for providing, maintaining, or modifying the services provided by a switching system.

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**OC**

Overload Control

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**OCI**

Overload Control Information

---

**OCS**

Online Charging System

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**OCX**

Operator Carrier Exchange

---

**ODB**

Operator Determined Barring

---

**OEM**

Original or Other Equipment Manufacturer

---

**OFCS**

Offline Charging System

---

**OID**

Object Identifier

---

**OLR**

Overload Control

---

**OM**

Operational Measurement

---

**OMA**

Open Mobile Alliance

---

**OOM**

Out Of Memory

---

**OOS**

Out Of Service

---

---

**Opex**

Operational Expenditure

---

**OS**

Operating System

---

**OSGi**

Open Services Gateway initiative

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**OSS/BSS**

Operational Support System/Business Support System

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**OSU**

Online Software Upgrade

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**OTDOA**

Observed Time Difference Of Arrival

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**OTT**

Over-the-top – On-line delivery of video and audio without the Internet service provider's involvement in the control or distribution of the content itself. Refers to content that arrives from a third party to the end user device, leaving the service provider responsible only for transporting IP packets.

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**OVA**

Open Virtualization Archive

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**OVS**

Open vSwitch – an open-source implementation of a distributed virtual multilayer switch providing a switching stack for hardware virtualization environments that supports multiple protocols/standards used in computer networks.

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**P**

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**P-CSCF**

Proxy Call Session Control Function

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**P-NIC**

Physical Network Interface Controller

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**P-TMSI**

Packet-Temporary Mobile Subscriber Identity

---

**P2PTV**

Peer-to-peer Television

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**PAT**

Port Address Translation

---

**PCB**

Printed Circuit Board

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<b>PCC</b>	Policy and Charging Control
<b>PCEF</b>	Policy and Charging Enforcement Function – Element that enforces the rules.
<b>PCI</b>	Peripheral Component Interconnect
<b>PCO</b>	Protocol Configuration Option
<b>PCP</b>	Priority Code Point
<b>PCRF</b>	Policy and Charging and Rules Function – Engine that manages the rules. Supports service data flow detection, policy enforcement, and flow-based charging.
<b>PCU</b>	Packet Control Unit
<b>PCUSN</b>	Packet Control Unit Support Node – A node in the BSS that interacts either directly or indirectly with all the BSS nodes, except the Transcoder Unit (TCU).
<b>PDCCP-SN</b>	Packet Data Convergence Protocol-Sequence Number
<b>PDHIR</b>	Packet Data Header Information Report
<b>PDN</b>	Packet Data Network – A communications system designed to carry packet data. Also Public Data Network
<b>PDP</b>	Packet Data Protocol/Policy Decision Point
<b>PDP Context</b>	Packet Data Protocol context – A packet data connection over which the UE and the network can exchange IP packets. The information sets held in the MS and the GSNs for a PDP address. It contains information such as PDP Type, QoS Profile, APN, and GGSN. Usage of these packet data connections is restricted to specific services, accessible via APNs.
<b>PDU</b>	Protocol Data Units – A data unit that is transferred among peer entities of a network and contains protocol information, such as control information and address information.
<b>PEC</b>	Product Engineering Code
<b>PEM</b>	Power Entry Module

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<b>PEP</b>	Policy Enforcement Point
<b>PF</b>	Physical Function
<b>PFC</b>	Packet Flow Context
<b>PFCP</b>	Packet Forwarding Control Protocol
<b>PFE</b>	Packet Forwarding Engine
<b>PFM</b>	Packet Flow Management
<b>PGW</b>	Packet Data Network Gateway
<b>PGW-C</b>	Packet Data Network Gateway - Control Plane
<b>PGW-U</b>	Packet Data Network Gateway - User Plane
<b>PHB</b>	Per-Hop Behavior
<b>PICMG</b>	PCI Industrial Computer Manufacturers Group
<b>PKI</b>	Public Key Infrastructure – A certificate-based mechanism for public key distribution.
<b>PLFP</b>	Partner Link Failure Propagation
<b>PLM</b>	Product Line Management
<b>PLMN</b>	Public Land Mobile Network – A GSM, GPRS, or UMTS public mobile communications network established and operated by an administration or its licensed operator(s), for the specific purpose of providing land mobile communication services to the public.
<b>PLS</b>	Pooling Licensing Server
<b>PM</b>	Performance Management
<b>PMC</b>	PCI Mezzanine Card

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**PMTU**

Path Maximum Transmission Unit

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**POJO**

Plain Old Java Object – Indicates that the object is not special, but is an ordinary Java Object (in particular is not an EJB).

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**PoP**

Point of Presence

---

**POST**

Power-On Self-Test

---

**PPP**

Point-to-Point Protocol

---

**PPS**

Packets Per Second

---

**PPV**

Pay-Per-View

---

**PS**

Packet Switched

---

**PSAP**

Public Safety Answering Points

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**PSK**

Pre-Shared Key – A form of IKE authentication.

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**PVI**

Pre-emption Vulnerability Indicator

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**PWS**

Public Warning System

---

**PXE**

Preboot eXecution Environment

---

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**Q**

---

**QCI**

QoS Class Identifier

---

**QEMU**

Quick Emulator

---

**QoE**

Quality of Experience

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**QoS**

Quality of Service

---

---

**QPI bus**

QuickPath Interconnect is a point-to-point processor interconnect that replaced the front-side bus (FSB) in certain desktop platforms.

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**QSFP+**

Quad Small Form-factor Pluggable Plus

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**QUIC**

Quick UDP Internet Connection

---

**QVOD**

Quasi Video on Demand

---

**R**

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**RA**

Router Advertisement

---

**RAB**

Radio Access Bearer

---

**RAC**

Routing Area Code

---

**RADIUS**

Remote Authentication (or Access) Dial-In User Service

---

**RAI**

Routing Area Identity

---

**RAI-FQDN**

Routing Area Identity-Fully Qualified Domain Name

---

**RAID**

Redundant Array of Independent Disks

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**RAM**

Random Access Memory

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**RAN**

Radio Access Network – The part of the mobile Telecommunication system located between the mobile phone and the core network.

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**RANAP**

Radio Access Network Application Part

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**RAT**

Radio Access Technology/Radio Access Type

---

**RAU**

Routing Area Update

---

**RE**

Routing Engine

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<b>REST</b>	Representational State Transfer
<b>REST/TCP</b>	Representational State Transfer Interface between SCEF and SCS/AS
<b>RFSP</b>	Radio Access Technology Frequency Selection Priority
<b>RGMII</b>	Reduced Gigabit Media-Independent Interface
<b>RHOSP</b>	Red Hat OpenStack Platform
<b>RIA</b>	Reporting Information Answer
<b>RIM</b>	Records and Information Management
<b>RIP</b>	Routing Information Protocol
<b>RIR</b>	Reporting Information Request
<b>RLC</b>	Radio Link Control
<b>RM</b>	Resource Manager
<b>RM VM</b>	Resource Manager Virtual Machine
<b>RMG</b>	Routing Monitoring Group
<b>RNC</b>	Radio Network Controller
<b>RoHS</b>	Restriction of Hazardous Substances Directive
<b>ROM</b>	Read-Only Memory
<b>RPC</b>	Reduced Partial CDR
<b>RPS</b>	Receive Packet Steering
<b>RRC</b>	Radio Resource Control



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**RS**

Router Solicitation

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**RSU**

Requested Service Unit

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**RSZ**

Regional Subscription Zone

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**RTM**

Rear Transition Module/Rear Termination Module

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**RTMP**

Real Time Messaging Protocol

---

**RTP**

Real Time Transport Protocol

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**RTSP**

Real Time Streaming Protocol

---

**RTT**

Real-Time Troubleshooting

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---

**S**

---

**S-CDR**

Serving GPRS Support Node - CDR

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**S-CSCF**

Serving Call Session Control Function

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**S-GW**

Serving Gateway

---

**S-NAPTR**

Straightforward-Naming Authority Pointer – limits the NAPTR record types to “a”, “s” and “”. S-NAPTR also simplifies the rewrite rules to replacement rules only.

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**S-SMO-CDR**

SGSN delivered Short message Mobile Originated – CDR

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**S-SMT-CDR**

SGSN delivered Short message Mobile Terminated – CDR

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**S1-AP**

S1 Application Protocol

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**S2b**

Interface between the ePDG and PGW (GTPv2) used to transport WLAN UE data from non-3GPP untrusted access networks through IPsec and GTP tunnels for secure data transport and connectivity to the PDN.

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<b>S6b</b>	Diameter-based interface between the 3GPP AAA Server and PGW used to manage address information, retrieve and update mobility-related information, authenticate APNs, and allocate UE IP addresses.
<b>S6m</b>	Interface between SCEF and HSS that supports subscriber information retrieval used to support device triggering.
<b>S6t</b>	Interface between SCEF and HSS
<b>SA</b>	Security Association – A unidirectional security agreement between two peers. Also System Architecture
<b>SaaS</b>	Software as a Service – A software delivery model in which software and its associated data are hosted centrally and are typically accessed by using a thin client, such as a Web browser over the Internet.
<b>SAD</b>	Security Association Database – A set of active Security Associations.
<b>SAE</b>	System Architecture Evolution
<b>SAE-GW</b>	Serving Gateway
<b>SAF</b>	Service Availability Forum
<b>SAI</b>	Service Area Identity
<b>SaMOG</b>	S2a Mobility based on GTP
<b>SAS</b>	SGSN Accounting Server – SGSN entity that provides SGSN billing functionality.
<b>SATA</b>	Serial Advanced Technology Attachment – High speed serial interface standard for storage devices.
<b>SC</b>	Subscriber Control: Primary purpose is to organize the related C-plane processes, such as GMM, SM, HLR Cache, and MAP Client.
<b>SCCP</b>	Signaling Connection Control Part
<b>SCEF</b>	Service Capability Exposure Function

---

<b>SCF</b>	Service Control Function
<b>SCG</b>	Secondary Cell Group
<b>SCP</b>	Service Control Point – External entity used to manage Prepaid SMS subscriber accounts.
<b>SCS</b>	Service Capability Server
<b>SCTP</b>	Stream Control Transmission Protocol
<b>SD</b>	Subscriber Data – Consists of the application handling the protocols involved in the 2G and 3G subscriber data path, namely LLC, SNDCP and GTP.
<b>SDC</b>	Service Data Container
<b>SDF</b>	Service Data Flow
<b>SDN</b>	Software-defined networking – An approach to building data networking equipment by decoupling the control plane from the data plane.
<b>SDR</b>	Sensor Data Record
<b>SDRAM</b>	Synchronous Dynamic Random Access Memory
<b>SECM</b>	Security Module
<b>SELV</b>	Safety Extra Low Voltage Circuits
<b>SeNB</b>	Secondary eNB
<b>SEND</b>	Secure Neighbor Discovery protocol
<b>SERDES</b>	Serializer/Deserializer
<b>Service Component</b>	A service model is defined using service components. A service component consists of a set of vendor objects across one or more vendors and their associations with other vendor objects for provisioning or deprovisioning a component, such as a packet filter, service, rule, etc. Service components can be designed to access multiple device instances using different API protocols. See also Association Service Component (page 3).

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**Service Component Instance**

An actual instance of a service component running in a lab or production network and derived from a service component.

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**Service Instance**

An actual instance of a service model running in a lab or production network. A service instance is derived from a service model and applied and rolled out to specific device instances in vTransactor.

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**Service Model**

Represents a service that consists of service components and their associations. The service model is a template used to create instances of the “service” which is then applied and rolled out to specific vendor object instances.

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**SFP**

Single Form-Factor Pluggable Transceiver/Small Form-Factor Pluggable

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**SFTP**

Secure FTP

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**SGMII**

Serial Gigabit Media Independent Interface

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**SGs**

3GPP Interface between MME and MSC/VLR.

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**SGSN**

Serving GPRS Support Node – Main functions are to detect new GPRS mobile stations in its service area, send/receive data packets to/from the mobile stations, and record the location of mobile stations inside its service area.

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**SGW**

Serving Gateway

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**SGW-C**

Serving Gateway - Control Plane

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**SGW-U**

Serving Gateway - User Plane

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**SH-BGP**

Single Hop Border Gateway Protocol

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**SIA**

Subscriber Information Answer

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**SIG**

Signaling or SS7/IP Gateway

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**SIG VM**

Signaling Virtual Machine – a type of VM on the MCC MME.

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**SIGTRAN**

Signaling Transport – Defines a family of protocols that provide reliable datagram transport service and user adaptation layer for traditional Signaling System 7-based protocols.

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<b>SIM</b>	Subscriber Identity Module
<b>SIR</b>	Subscriber Information Request
<b>SLB</b>	Steering Load Balancer – a type of VM for the MCC MME. Also Session Load Balancer
<b>SLF</b>	Subscriber Locator Function
<b>SLg</b>	Diameter interface between the MME and GMLC.
<b>SMPP</b>	Short Message Peer-to-Peer Protocol
<b>SMS</b>	Short Messages Service – Provides the means to transfer messages between a GPRS PLMN MS and a Short Message entity through a service center.
<b>SMSC</b>	Short Message Service Center
<b>SMTP</b>	Simple Mail Transfer Protocol
<b>Snapshot</b>	A version of a service instance in vTransactor. When you make changes and updates to a specified service instance and then click the Snapshot icon, vTransactor creates a new snapshot of that service instance. If you make one additional change and do not click the Snapshot icon, the changes will be pending and will not be part of the latest snapshot (until you click the Snapshot icon). Snapshots can be compared with devices (Config Diff) and can be rolled out to a selected Footprint. See also Footprint (page 11).
<b>SNDCP</b>	Sub Network Dependent Convergence Protocol – A layer 3 protocol responsible for segmentation or reassembly and possibly compression of data packets for transport between the MS and the SGSN.
<b>SNI</b>	Server Name Indication
<b>SNMP</b>	Simple Network Management Protocol – Used by network management systems to monitor network-attached devices for conditions requiring administrative attention.
<b>SNOW3G</b>	A stream cipher that uses a 128-bit key and IV.
<b>SNR</b>	Seamless National Roaming

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**SNS**

Sub-Network Services

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**SOAP**

Simple Object Access Protocol – protocol used for exchanging XML-based messages over computer network, normally using HTTP/HTTPS. SOAP forms the foundation layers of the Web Services stack, providing a basic messaging framework.

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**SON**

Self Optimized Networks

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**southbound**

Interfaces from function under discussion towards the 3GPP network functions.

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**SPA**

Security Policy Agent

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**SPD**

Security Policy Database Also Rule set used by an Ipsec implementation to secure traffic

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**SPI**

Security Parameter Index/Serial Peripheral Interface

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**SPM**

Security Policy Manager

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**SPR**

Subscriber Profile Repository

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**SR-IOV**

Single Root I/O Virtualization

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**SRAM**

Static Random Access Memory

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**SRNS**

Serving Radio Network Subsystem

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**SRV**

Service records – a specification of data in the Domain Name System defining the location (hostname and port number) of servers for specified services defined in RFC 2782.

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**SS7**

Signaling System 7 – Common channel signaling system designed for the control of voice and non-voice services and for use in a digital environment in which signaling information may be sent at 64 kbit/s.

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**SSD**

Solid State Drive – An all-electronic storage device that is an alternative to a hard disk drive. Used in smartphones, MP3 players, digital cameras and laptop computers, solid state drives are faster than hard disks because there is zero latency (no read/write head to move).

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**SSH**

Secure Shell

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**SSM**

Subscriber Services Module – A cryptographic network protocol for operating network services securely over an unsecured network.

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**SSO**

Single Sign On – Provides single authentication/sign-on for multiple applications.

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**STa**

Interface between the non-3GPP access network and the 3GPP AAA server or between a non-3GPP access network and the 3GPP AAA proxy.

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**STA**

Session-Termination-Answer Diameter command code

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**STI**

Standard Transcoding Interface

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**STP**

Shielded Twisted Pair

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**STPx**

System Test Phase <x> (milestone)

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**STR**

Session Termination Request

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**SU**

Service Unit

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**SVOD**

Subscription Video on Demand

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**SWm**

Interface linking the 3GPP AAA server and the ePDG or between the 3GPP AAA Proxy and the ePDG.

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**SWu**

Interface between the ePDG and 3GPP Diameter-based AAA Server used to authenticate UE through EAP and IKEv2 signaling, secure data transport through IPsec tunnels, and IP address allocation to UEs. This interface is from the ePDG to UE in non-3GPP untrusted IP access networks. It supports roaming/mobility, tunnel authentication and authorization of subscribers.

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**SX**

Short Range Optical Transceiver

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**T**

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**T-PDU**

Tunneled PDU

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<b>T4</b>	Interface between SCEF and SMS-SC that supports the transfer of device trigger messages and the success/failure reporting of device trigger deliveries to the UE.
<b>T6a</b>	Interface between SCEF and MME
<b>T8</b>	Interface between SCEF and SCS/AS
<b>TA</b>	Tracking Area
<b>TAC</b>	Tracking Area Code
<b>TACACS+</b>	Terminal Access Controller Access-Control System Plus
<b>TAI</b>	Tracking Area Identity
<b>TAM</b>	Traffic Anchoring Mode
<b>TAS</b>	Technical Assistance Services
<b>TAU</b>	Tracking Area Update
<b>TBD</b>	To Be Determined
<b>TCAP</b>	Transaction Capabilities Application Part – TCAP uses the MTP and SCCP software layers to provide connectionless signaling for transaction services. Simply put, it allows two nodes to communicate with each other in the same language.
<b>TCE</b>	Trace Collection Entity
<b>TCP</b>	Transmission Control Protocol
<b>TCP/IP</b>	Transmission Control Protocol/Internet protocol – A protocol stack - designed to connect different networks - on which the Internet is based.
<b>TDF</b>	Traffic Detection Function
<b>TDM</b>	Time Division Multiplex



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<b>TDP</b>	Trigger Detection Point
<b>TEID</b>	Tunnel End Identifier
<b>TFD</b>	Toll Free Data
<b>TFT</b>	Traffic Flow Template
<b>TI</b>	Technology Introduction
<b>TIPC</b>	Transparent Inter Process Communication
<b>TLLI</b>	Temporary Logical Link Identifier – Describes an identified MS registered in the network and owning a LLC context with the SGSN.
<b>TLS</b>	Transport Layer Security
<b>TLV</b>	Type Length Value
<b>TODA</b>	Time Of Day Accounting
<b>TOS</b>	Type of Service/Terms of Service
<b>TPE</b>	Twisted-Pair Ethernet
<b>TPS</b>	Transactions Per Second
<b>TTCT</b>	Tariff Time Change Trigger
<b>TUI</b>	text user interface
<b>TVOD</b>	Transactional Video on Demand
<b>TWAG</b>	Trusted WLAN Access Gateway
<b>TWAP</b>	Trusted WLAN AAA Proxy

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## U

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### **U-plane**

User plane – Function associated with the transfer of actual user data to and from a PDN.

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### **UAProf**

User Agent Profile

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### **UART**

Universal Asynchronous Receiver / Transmitter

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### **UBR**

Unspecified Bit Rate

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### **UDP**

User Datagram Protocol

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### **UDP/IP**

User Datagram Protocol/Internet Protocol – A transport layer, connectionless mode protocol, providing a datagram mode of communication for delivery of packets to a remote or local user.

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### **UE**

User Equipment – Device used directly by an end-user to communicate in UMTS and LTE.

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### **UE-AMBR**

User Equipment – Aggregated Maximum Bit Rate

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### **UGL**

Update GPRS Location

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### **UIM**

User Identity Module

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### **UL**

Underwriters Laboratories

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### **ULI**

User Location Information

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### **UMTS**

Universal Mobile Telecommunications System

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### **UP**

User Plane – See U-Plane.

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### **UPF**

User Plane Function

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### **UPM**

User Datagram Protocol (UDP) Path Manager – A software component on the MME.

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### **UPSM**

User Datagram Protocol (UDP) Path Steering Manager

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**URI**

Universal Resource Indicator

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**USD**

UMTS Subscriber Data

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**USIM**

UMTS SIM

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**USU**

Used Service Units

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**UTC**

Universal Time Coordinated

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**UTRA**

Universal Terrestrial Radio Access

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**UTRAN**

Universal Terrestrial Radio Access Network

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**V**

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**VAS**

Value-Added Service

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**VASP**

Value-Added Service Provider

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**VBMC**

Virtual Baseboard Management Controller

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**VBR**

Variable Bit Rate – The bit rate used in sound or video encoding. VBR files vary the amount of output data per time segment. VBR allows a higher bit rate (and therefore more storage space) to be allocated to the more complex segments of media files while less space is allocated to less complex segments.

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**vBuilder**

An ASAP tool that works in a carrier's corporate network to construct service orchestration templates or recipes that perform management operations on any class of network device or application using a connector type such as YANG or REST.

---

**vCPU**

Virtual Central Processing Unit is assigned to each VM in a cloud environment.

---

**vDISK (or VD)**

Virtual Disk – a virtual device that provides an area of usable storage capacity on one or more physical disk drive(s) in a computer system.

---

**Vendor Component**

A vendor component represents a collection of vendor objects to be applied on a device type. A vendor component is associated with only one device type and one API protocol

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type. A service component may have several vendor components.

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**Vendor Object**

A device type may have one or more vendor objects used in a service model and based on an API protocol or CLI. Typically, vendor objects are parsed from the vendor schema derived from the NETCONF/YANG or REST API. A subject matter expert (SME) may also create the vendor objects manually, for example, for CLI (SSH). A SME may configure a specific vendor object of a device type with parameters or hard code the values of the parameters.

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**Vendor Object Instance**

One or more instances of a vendor object contained by any number of service instances.

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**Vendor Schema Object**

Refers to a supported API protocol (e.g., YANG or REST) to import and on which to base and create vendor objects in vBuilder.

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**vEPC**

virtualized Evolved Packet Core – a mobile core network system that accommodates LTE access systems.

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**vePDG**

virtualized Evolved Packet Data Gateway

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**VIM**

Virtualized Infrastructure Manager

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**VLAN**

Virtual Local Area Network

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**VLR**

Visitor Location Register

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**VM**

Virtual Machine

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**vMCM**

Virtual Management Configuration Module – A VM instance that runs core processes, including node and configuration functions.

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**vMemory**

Virtual Memory – a container for managing an array of memory containers.

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**VMM**

Virtual Machine Management

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**VNF**

Virtual Network Function – Software implementation of a network function that can be deployed on an NFVI.

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**vNIC**

Virtualized Network Interface Card – a vNIC is a VM that provides virtual network interface by defining a range of MAC addresses.

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**VOD**

Video On Demand

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**VPLMN**

Visited Public Land Mobile Network

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**VPN**

Virtual Private Network

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**VRF**

Virtual Routing and Forwarding

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**VSA**

Vendor-Specific Attribute

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**vSSF**

virtual Slice Selection Function

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**vSwitch**

Virtual Switch – a software application that allows communication between VMs.

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**vTransactor**

An ASAP tool installed in customer NOCs to implement the service orchestration recipes in a carrier's network.

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**W**

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**WAE**

Wireless Application Environment

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**WAG**

Wireless Access Gateway

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**WAP**

Wireless Application Protocol

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**WCDMA**

Wideband Code Division Multiple Access

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**WDP**

Wireless Datagram Protocol

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**WEEE**

Waste from Electrical and Electronic Equipment

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**WLAN**

Wireless Local Area Network

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**WP-HTTP**

Wireless Profiled HTTP

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**WP-TCP**

Wireless Profiled TCP

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**WSM**

Workflow Services Module – A VM instance that provides workflow services.

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**WSP**

Wireless Session Protocol

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**WTLS**

Wireless Transport Layer Security

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**WTP**

Wireless Transaction Protocol

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**X**

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**X.509**

A specific implementation of PKI using trusted certificates.

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**XML**

Extensible Markup Language

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